

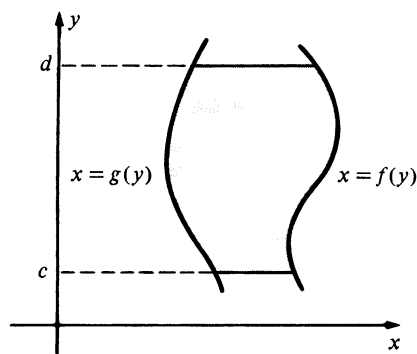


FIGURE 6.7

Sometimes it is necessary to find the area A of the region bounded by the graphs of $y = c$ and $y = d$, and of two equations of the form $x = f(y)$ and $x = g(y)$, where f and g are continuous functions and $f(y) \geq g(y)$ for all y in $[c, d]$ (see Figure 6.7). In a manner similar to our earlier discussion, but with the roles of x and y interchanged, we obtain the formula

$$A = \int_c^d [f(y) - g(y)] dy$$

where we now regard y as the independent variable. We shall refer to this technique as **integration with respect to y** , whereas Theorem (6.1) is called **integration with respect to x** .

Summation techniques can also be applied to regions of the type shown in Figure 6.7. In this case we select points on the y -axis with y -coordinates $y_0, y_1, \dots, y_n$, where $y_0 = c$ and $y_n = d$, thereby obtaining a partition of the interval $[c, d]$ into subintervals of width $\Delta y_i = y_i - y_{i-1}$. For each i we choose a number w_i in $[y_{i-1}, y_i]$ and consider rectangles that have areas $[f(w_i) - g(w_i)] \Delta y_i$, as illustrated in Figure 6.8. This leads to

$$A = \lim_{\|P\| \rightarrow 0} \sum_i [f(w_i) - g(w_i)] \Delta y_i = \int_c^d [f(y) - g(y)] dy,$$

where the last equality follows from the definition of the definite integral.

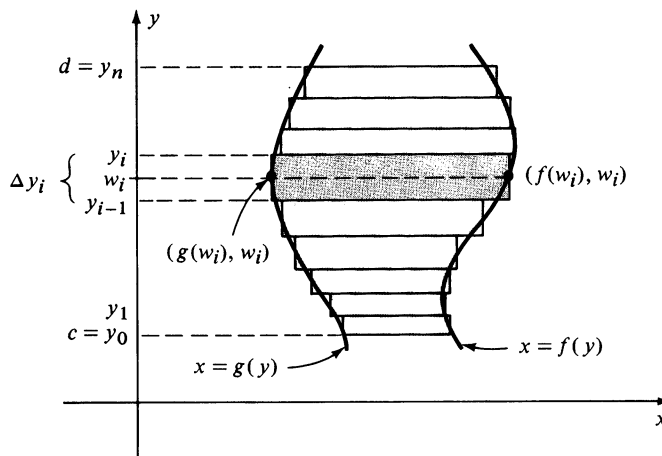


FIGURE 6.8

Example 4 Find the area of the region bounded by the graphs of the equations $2y^2 = x + 4$ and $x = y^2$.

Solution Two sketches of the region are shown in Figure 6.9, where (i) illustrates the situation that occurs if we use vertical rectangles (integration with respect to x), and (ii) is the case if we use horizontal rectangles (integration with respect to y).

Referring to (i) of the figure we see that several definite integrals are required to find the area. (Why?) However, in (ii) we can use integration

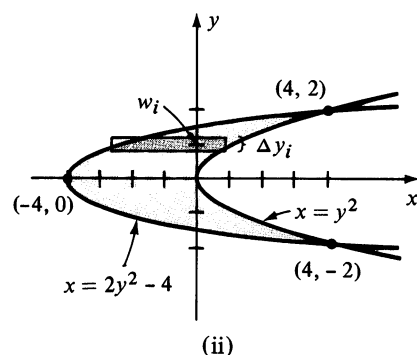
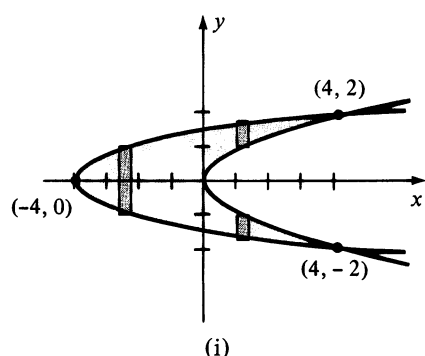


FIGURE 6.9

with respect to y to find the area with only one integration. Letting $f(y) = y^2$, $g(y) = 2y^2 - 4$, and referring to (ii) of Figure 6.9, the length $f(w_i) - g(w_i)$ of a horizontal rectangle is

$$w_i^2 - (2w_i^2 - 4).$$

Since the width is Δy_i , the area of the rectangle is

$$[w_i^2 - (2w_i^2 - 4)] \Delta y_i.$$

Hence, the area of R is

$$\begin{aligned} A &= \lim_{\|P\| \rightarrow 0} \sum_i [w_i^2 - (2w_i^2 - 4)] \Delta y_i \\ &= \int_{-2}^2 [y^2 - (2y^2 - 4)] dy \\ &= \int_{-2}^2 (4 - y^2) dy \\ &= \left[4y - \frac{y^3}{3} \right]_{-2}^2 = \left[8 - \frac{8}{3} \right] - \left[-8 - \left(-\frac{8}{3} \right) \right] = \frac{32}{3}. \end{aligned}$$

Actually, the integration could have been further simplified. For example, since the x -axis bisects the region, it is sufficient to find the area of that part of the region that lies above the x -axis and double it, obtaining

$$A = 2 \int_0^2 [y^2 - (2y^2 - 4)] dy. \quad \blacksquare$$

Throughout this section we have assumed that the graphs of the functions (or equations) do not cross one another in the interval under discussion. For example, in Theorem (6.1) we demanded that $f(x) \geq g(x)$ for all x in $[a, b]$. If the graphs of f and g cross at one point $P(c, d)$, where $a < c < b$, and we wish to find the area bounded by the graphs from $x = a$ to $x = b$, then the theory developed in this section may still be used; however, *two* integrations are required, one corresponding to the interval $[a, c]$ and the other to $[c, b]$. If the graphs cross *several* times, then several integrals may be necessary. Problems whose graphs cross one or more times appear in Exercises 33–36.

In scientific investigations, physical interpretations are often attached to areas. One illustration of this occurs in the *theory of elasticity* where, to test the strength of a material, an investigator records values of strain that correspond to different loads (stresses). The sketch in Figure 6.10 is a typical stress-strain diagram for a sample of an elastic material such as vulcanized rubber. Note that it is customary to assign stress values in the vertical direction.

Referring to the figure, we see that as the load applied to the material (the stress) was increased, the strain (indicated by the arrows on the solid part of the graph) increased until the material stretched to five times its original length. As the load was decreased, the elastic material returned to its original length; however, the same graph was not retraced. Instead, the

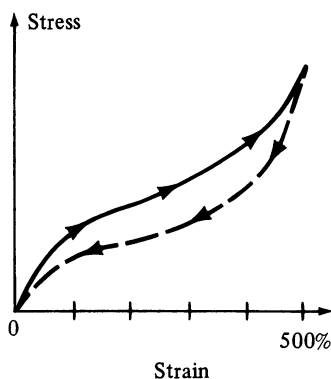


FIGURE 6.10 Stress-strain diagram for an elastic material

graph indicated by the dashes was obtained. This phenomenon is called *elastic hysteresis*. (A similar occurrence takes place in the study of magnetic materials, where it is called *magnetic hysteresis*.) The two curves in the figure make up a *hysteresis loop* for the material. It can be shown that the area of the region enclosed by this loop is numerically equal to the energy dissipated within the elastic (or magnetic) material during the test. In the case of vulcanized rubber, the larger the area, the better the material is for absorbing vibrations.

6.1 Exercises

In Exercises 1–20 sketch the region bounded by the graphs of the given equations, show a typical vertical or horizontal rectangle, and find the area of the region.

- 1 $y = 1/x^2, y = -x^2, x = 1, x = 2$
- 2 $y = \sqrt{x}, y = -x, x = 1, x = 4$
- 3 $y^2 = -x, x - y = 4, y = -1, y = 2$
- 4 $x = y^2, y - x = 2, y = -2, y = 3$
- 5 $y = x^2 + 1, y = 5$
- 6 $y = 4 - x^2, y = -4$
- 7 $y = x^2, y = 4x$
- 8 $y = x^3, y = x^2$
- 9 $y = 1 - x^2, y = x - 1$
- 10 $x + y = 3, y + x^2 = 3$
- 11 $y^2 = 4 + x, y^2 + x = 2$
- 12 $x = y^2, x - y - 2 = 0$
- 13 $y = x, y = 3x, x + y = 4$
- 14 $x - y + 1 = 0, 7x - y - 17 = 0, 2x + y + 2 = 0$
- 15 $y = x^3 - x, y = 0$
- 16 $y = x^3 - x^2 - 6x, y = 0$
- 17 $x = 4y - y^3, x = 0$
- 18 $x = y^{2/3}, x = y^2$
- 19 $y = x\sqrt{4 - x^2}, y = 0$
- 20 $y = x\sqrt{x^2 - 9}, y = 0, x = 5$
- 21 Find the area of the region between the graphs of the equations $y = \sin 4x$ and $y = 1 + \cos \frac{1}{3}x$ from $x = 0$ to $x = \pi$.
- 22 Find the area of the region between the graphs of the equations $y = 4 + \cos 2x$ and $y = 3 \sin \frac{1}{2}x$ from $x = 0$ to $x = \pi$.

- 23 If R is the region bounded by the graph of $(x - 4)^2 + y^2 = 9$, express the area A of R as a limit of a sum. Find A without integrating.

- 24 If R is the region bounded by the graphs of $2x + 3y = 6$, $x = 0$, and $y = 0$, express the area A of R as a limit of a sum. Find A without integrating.

Each of Exercises 25–32 represents a limit of a sum for a function f on the indicated interval. Interpret each limit as an area of a region R and find its value. Sketch the region R .

- 25 $\lim_{\|P\| \rightarrow 0} \sum_i (4w_i + 1) \Delta x_i; [0, 1]$
- 26 $\lim_{\|P\| \rightarrow 0} \sum_i (w_i - w_i^3) \Delta x_i; [0, 1]$
- 27 $\lim_{\|P\| \rightarrow 0} \sum_i (4 - w_i^2) \Delta y_i; [0, 1]$
- 28 $\lim_{\|P\| \rightarrow 0} \sum_i \sqrt{3w_i + 1} \Delta y_i; [0, 1]$
- 29 $\lim_{\|P\| \rightarrow 0} \sum_i [w_i/(w_i^2 + 1)^2] \Delta x_i; [2, 5]$
- 30 $\lim_{\|P\| \rightarrow 0} \sum_i (5w_i/\sqrt{9 + w_i^2}) \Delta x_i; [1, 4]$
- 31 $\lim_{\|P\| \rightarrow 0} \sum_i [(5 + \sqrt{w_i})/\sqrt{w_i}] \Delta y_i; [1, 4]$
- 32 $\lim_{\|P\| \rightarrow 0} \sum_i w_i^{-2} \Delta y_i; [-5, -2]$

In Exercises 33–36 find the area of the region that lies between the graphs of f and g when x is restricted to the given interval $[a, b]$. (Hint: $|f(x) - g(x)|$ is the distance between the graphs when x is in $[a, b]$.)

- 33 $f(x) = 6 - 3x^2, g(x) = 3x; [0, 2]$
- 34 $f(x) = x^2 - 4, g(x) = x + 2; [1, 4]$
- 35 $f(x) = \sin x, g(x) = \cos x; [0, 2\pi]$
- 36 $f(x) = x^2, g(x) = x^3; [-1, 2]$
- 37 The shape of a particular stress-strain diagram is shown in the accompanying figure (refer to the last paragraph of

this section). Estimate y -coordinates and approximate the area of the region enclosed by the hysteresis loop using (with $n = 6$) (a) the Trapezoidal Rule; (b) Simpson's Rule.

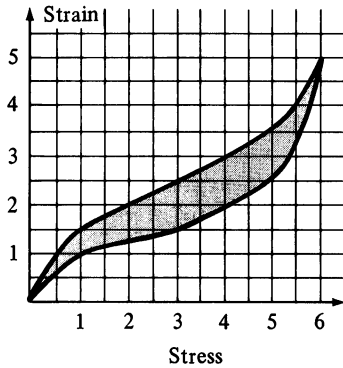


FIGURE FOR EXERCISE 37

- 38 Suppose the functional values of f and g in the following table were obtained empirically. Assuming that f and g are continuous, approximate the area between their graphs from $x = 1$ to $x = 5$ using (with $n = 8$) (a) the Trapezoidal Rule; (b) Simpson's Rule.

x	1	1.5	2	2.5	3	3.5	4	4.5	5
$f(x)$	3.5	2.5	3	4	3.5	2.5	2	2	3
$g(x)$	1.5	2	2	1.5	1	0.5	1	1.5	1

6.2 Solids of Revolution

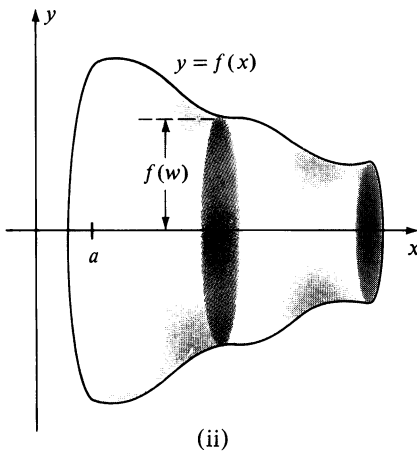
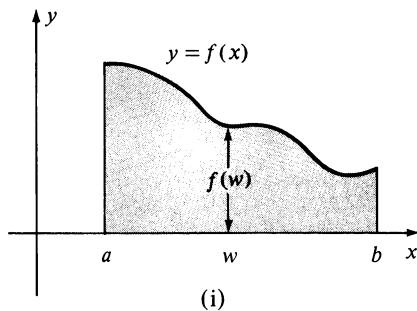


FIGURE 6.11

The volume of an object plays an important role in many problems that arise in the physical sciences. For example, it is essential to know the volume in order to find the center of gravity or moment of inertia of a homogeneous solid. (These concepts will be discussed later in the text.) The task of determining the volume of an irregularly shaped object is often difficult, if not impossible. For this reason, we shall begin with objects that have simple shapes. Included in this category are the solids of revolution discussed here and in the next section.

If a region in a plane is revolved about a line in the plane, the resulting solid is called a **solid of revolution**, and the solid is said to be **generated** by the region. The line about which the revolution takes place is called an **axis of revolution**. If the region bounded by the graph of a continuous nonnegative valued function f , the x -axis, and the graphs of $x = a$ and $x = b$ (see (i) of Figure 6.11) is revolved about the x -axis, a solid of the type shown in (ii) of Figure 6.11 is generated. For example, if f is a constant function, then the region is rectangular and the solid generated is a right circular cylinder. If the graph of f is a semicircle with endpoints of a diameter at the points $(a, 0)$ and $(b, 0)$ where $b > a$, then the solid of revolution is a sphere with diameter $b - a$. If the given region is a right triangle with base on the x -axis and two vertices at the points $(a, 0)$ and $(b, 0)$ with the right angle at one of these points, then a right circular cone is generated.

If a plane perpendicular to the x -axis intersects the solid shown in (ii) of Figure 6.11, a circular cross section is obtained. If, as indicated in the figure, the plane passes through the point on the x -axis with x -coordinate w , then the radius of the circle is $f(w)$ and hence its area is $\pi[f(w)]^2$. We shall arrive at a definition for the volume of such a solid of revolution by using

Riemann sums in a manner similar to that used for areas in the previous section.

Suppose f is continuous and $f(x) \geq 0$ for all x in $[a, b]$. Consider a Riemann sum $\sum_i f(w_i) \Delta x_i$, where w_i is any number in the i th subinterval $[x_{i-1}, x_i]$ of a partition P of $[a, b]$. Geometrically, this gives us a sum of areas of rectangles of the types shown in (i) of Figure 6.12. The solid generated by the polygon formed by these rectangles has the appearance shown in (ii) of the figure. Observe that the i th rectangle generates a circular disc (that is, a “flat” right circular cylinder) of base radius $f(w_i)$ and altitude, or “thickness,” $\Delta x_i = x_i - x_{i-1}$. The volume of this disc is the area of the base times the altitude, that is, $\pi[f(w_i)]^2 \Delta x_i$. The sum of the volumes of all such discs is the volume of the solid shown in (ii) of Figure 6.12 and is given by

$$\sum_i \pi[f(w_i)]^2 \Delta x_i.$$

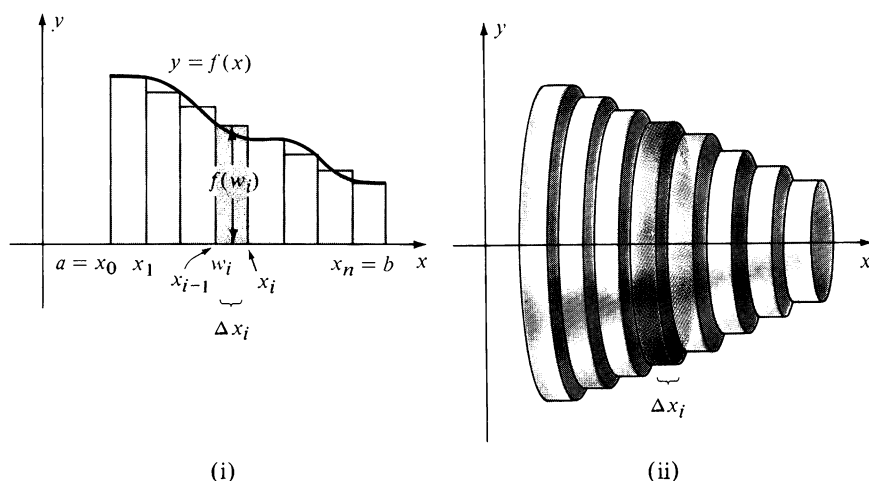


FIGURE 6.12

The sum may be regarded as a Riemann sum for the function h defined by $h(x) = \pi[f(x)]^2$. Intuitively, it appears that if $\|P\|$ is close to zero, then the sum is close to the volume of the solid. It is natural, therefore, to define the volume of revolution as the limit of this sum as follows.

Definition (6.3)

Let f be continuous on $[a, b]$. The **volume** V of the solid of revolution generated by revolving the region bounded by the graphs of f , $x = a$, $x = b$, and the x -axis about the x -axis is

$$V = \lim_{\|P\| \rightarrow 0} \sum_i \pi[f(w_i)]^2 \Delta x_i = \int_a^b \pi[f(x)]^2 dx.$$

The fact that the limit of the sum in Definition (6.3) equals $\int_a^b \pi[f(x)]^2 dx$ follows from the definition of the definite integral. Hereafter, when considering

limits of Riemann sums, the meanings of all symbols will not be explicitly pointed out. Instead, it will be assumed that the reader is aware of the significance of symbols such as $\|P\|$, w_i , and Δx_i .

The requirement that $f(x) \geq 0$ for all x in $[a, b]$ was omitted in Definition (6.3). If f is negative for some x , as illustrated in (i) of Figure 6.13, and if the region bounded by the graphs of f , $x = a$, $x = b$, and the x -axis is revolved about the x -axis, a solid of the type shown in (ii) of the figure is obtained. This solid is the same as that generated by revolving the region under the graph of $y = |f(x)|$ from a to b about the x -axis. Since $|f(x)|^2 = [f(x)]^2$, the limit in Definition (6.3) is the desired volume.

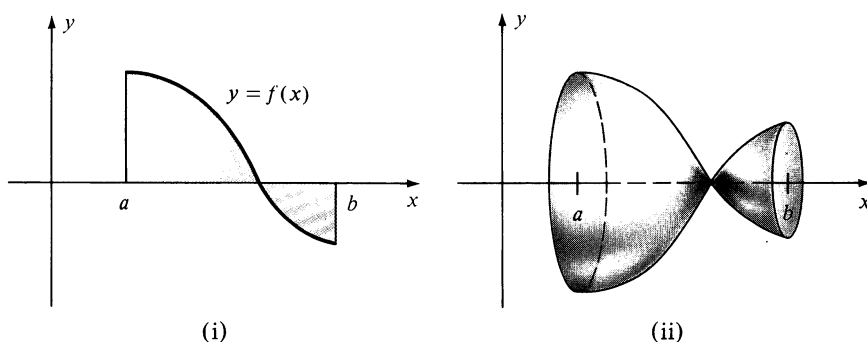


FIGURE 6.13

Example 1 If $f(x) = x^2 + 1$, find the volume of the solid generated by revolving the region under the graph of f from -1 to 1 about the x -axis.

Solution The solid is illustrated in Figure 6.14. Included in the sketch is a typical rectangle and the disc that it generates. Since the radius of the disc is $w_i^2 + 1$, its volume is

$$\pi(w_i^2 + 1)^2 \Delta x_i$$

and, as in (6.3),

$$\begin{aligned} V &= \lim_{\|P\| \rightarrow 0} \sum_i \pi(w_i^2 + 1)^2 \Delta x_i \\ &= \int_{-1}^1 \pi(x^2 + 1)^2 dx = \pi \int_{-1}^1 (x^4 + 2x^2 + 1) dx \\ &= \pi \left[\frac{1}{5}x^5 + \frac{2}{3}x^3 + x \right]_{-1}^1 \\ &= \pi \left[\left(\frac{1}{5} + \frac{2}{3} + 1 \right) - \left(-\frac{1}{5} - \frac{2}{3} - 1 \right) \right] = \frac{56}{15} \pi. \end{aligned}$$

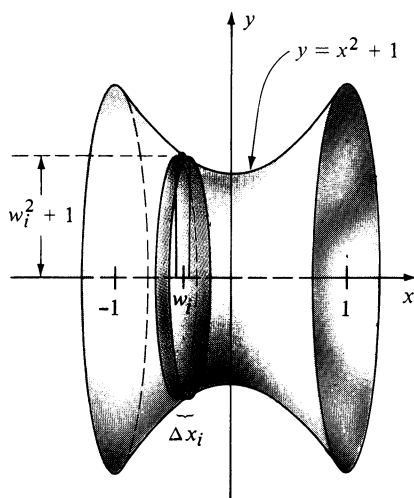


FIGURE 6.14

If we had used the symmetry of the graph in Example 1, then the volume could have been calculated by integrating from 0 to 1 and doubling the result. Another solution consists of substitution of $x^2 + 1$ for $f(x)$ in the

formula $V = \int_a^b \pi[f(x)]^2 dx$. We shall not ordinarily specify the units of measure. If the unit of linear measurement is inches, the volume is expressed in cubic inches. If x is in cm, then V is in cm^3 , etc.

Sometimes it is convenient to find volumes by integrating with respect to y . For example, consider a region bounded by horizontal lines with y -intercepts c and d , by the y -axis, and by the graph of $x = g(y)$, where the function g is continuous for all y in $[c, d]$. If this region is revolved about the y -axis, the volume V of the resulting solid may be found by interchanging the roles of x and y in Definition (6.3). Specifically, let P be the partition of the interval $[c, d]$ determined by the numbers $c = y_0, y_1, \dots, y_n = d$. Let $\Delta y_i = y_i - y_{i-1}$, let w_i be any number in the i th subinterval, and consider the rectangles of length $g(w_i)$ and width Δy_i as illustrated in (i) of Figure 6.15. The solid generated by revolving these rectangles about the y -axis is illustrated in (ii) of the figure.

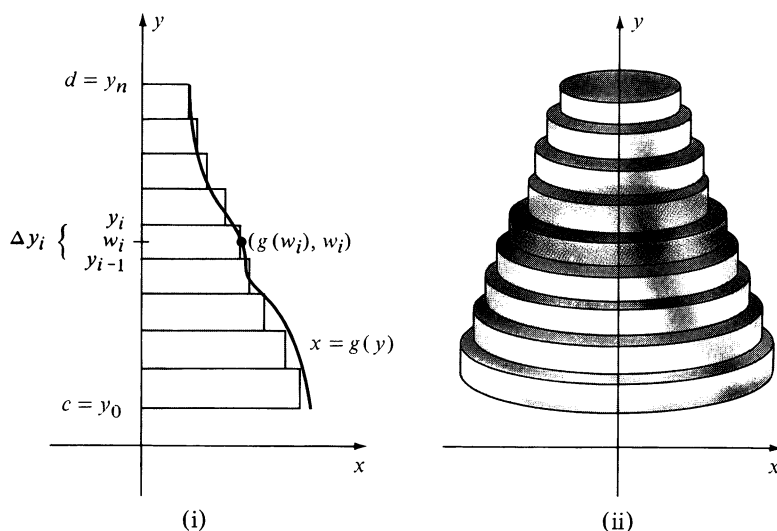


FIGURE 6.15

The volume of the disc generated by the i th rectangle is $\pi[g(w_i)]^2 \Delta y_i$. Summing and taking the limit gives us the following analogue of Definition (6.3).

Definition (6.4)

Let g be continuous on $[c, d]$. The **volume** V of the solid of revolution generated by revolving the region bounded by the graphs of $x = g(y)$, $y = c$, $y = d$, and the y -axis about the y -axis is

$$V = \lim_{\|P\| \rightarrow 0} \sum_i \pi[g(w_i)]^2 \Delta y_i = \int_c^d \pi[g(y)]^2 dy.$$

Example 2 The region bounded by the y -axis and the graphs of $y = x^3$, $y = 1$, and $y = 8$ is revolved about the y -axis. Find the volume of the resulting solid.

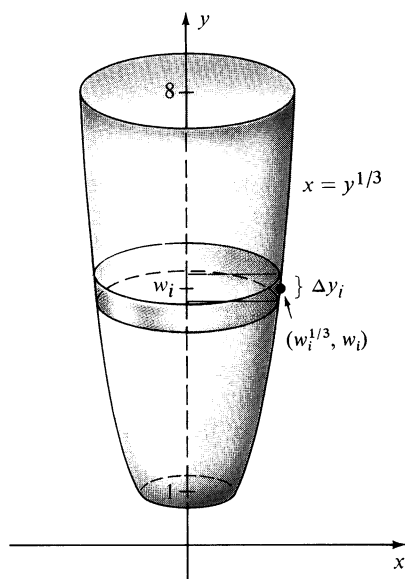


FIGURE 6.16

Solution The solid is sketched in Figure 6.16 together with a disc generated by a typical horizontal rectangle. Since we plan to integrate with respect to y , we solve the equation $y = x^3$ for x in terms of y , obtaining $x = y^{1/3}$. If we let $x = g(y) = y^{1/3}$, then as shown in Figure 6.16, the radius of a typical disc is $g(w_i) = w_i^{1/3}$ and its volume is $\pi(w_i^{1/3})^2 \Delta y_i$. Applying Definition (6.4) with $g(y) = y^{1/3}$ gives us

$$\begin{aligned} V &= \lim_{\|P\| \rightarrow 0} \sum_i \pi(w_i^{1/3})^2 \Delta y_i \\ &= \int_1^8 \pi(y^{1/3})^2 dy = \pi \int_1^8 y^{2/3} dy \\ &= \pi \left(\frac{3}{5} \right) \left[y^{5/3} \right]_1^8 = \frac{3}{5} \pi [8^{5/3} - 1] = \frac{93}{5} \pi. \end{aligned}$$

Let us next consider a region bounded by the graphs of $x = a$, $x = b$, and of two continuous functions f and g where $f(x) \geq g(x) \geq 0$ for all x in $[a, b]$. If this region is revolved about the x -axis, a solid of the type illustrated in (i) of Figure 6.17 may be obtained. Note that if $g(x) > 0$ for all x in $[a, b]$, then there is a hole through the solid.

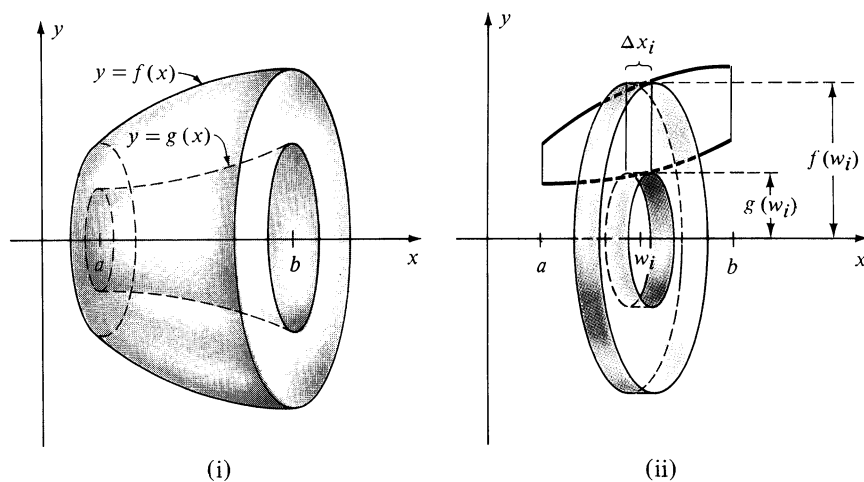


FIGURE 6.17

The volume V may be found by subtracting the volume of the solid generated by the smaller region from the volume of the solid generated by the larger region. Using Definition (6.3) gives us

$$V = \int_a^b \pi[f(x)]^2 dx - \int_a^b \pi[g(x)]^2 dx = \int_a^b \pi\{[f(x)]^2 - [g(x)]^2\} dx.$$

The last integral has an interesting interpretation as a limit of a sum. As illustrated in (ii) of Figure 6.17, a rectangle extending from the graph of g

to the graph of f , through the points with x -coordinate w_i , generates a washer-shaped solid whose volume is

$$\pi[f(w_i)]^2 \Delta x_i - \pi[g(w_i)]^2 \Delta x_i = \pi\{[f(w_i)]^2 - [g(w_i)]^2\} \Delta x_i.$$

Summing the volumes of all such washers and taking the limit gives us the desired integral formula. When working problems of this type it is often convenient to use the following general formula:

(6.5) Volume of a washer = $\pi[(\text{outer radius})^2 - (\text{inner radius})^2] \cdot (\text{thickness})$.

In integration problems the thickness will be given by either Δx_i or Δy_i . To find the volume we take a limit of a sum of volumes of these washers.

Example 3 The region bounded by the graphs of the equations $x^2 = y - 2$, $2y - x - 2 = 0$, $x = 0$, and $x = 1$ is revolved about the x -axis. Find the volume of the resulting solid.

Solution The region and a typical rectangle are sketched in (i) of Figure 6.18. Since we wish to integrate with respect to x we solve the first two equations for y in terms of x , obtaining $y = x^2 + 2$ and $y = \frac{1}{2}x + 1$. The washer generated by the rectangle in (i) is illustrated in (ii) of Figure 6.18.

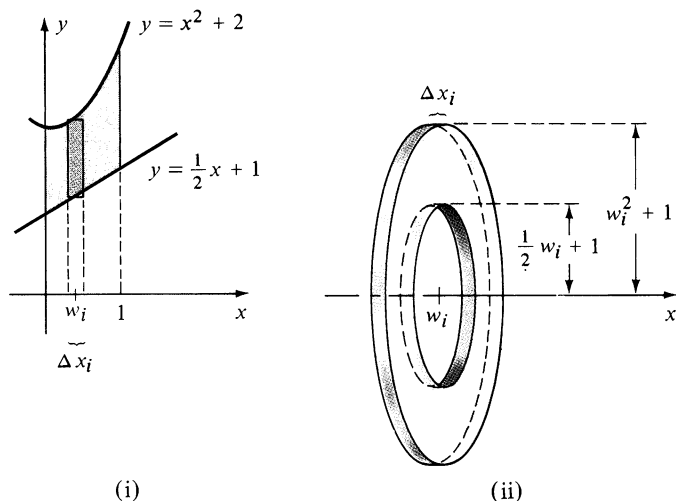


FIGURE 6.18

Since the outer radius of the washer is $w_i^2 + 2$ and the inner radius is $\frac{1}{2}w_i + 1$, its volume (see (6.5)) is

$$\pi \left[(w_i^2 + 2)^2 - \left(\frac{1}{2}w_i + 1 \right)^2 \right] \Delta x_i.$$

Taking the limit of a sum of such volumes gives us

$$\begin{aligned}
 V &= \int_0^1 \pi \left[(x^2 + 2)^2 - \left(\frac{1}{2}x + 1 \right)^2 \right] dx \\
 &= \pi \int_0^1 \left(x^4 + \frac{15}{4}x^2 - x + 3 \right) dx \\
 &= \pi \left[\frac{1}{5}x^5 + \frac{5}{4}x^3 - \frac{1}{2}x^2 + 3x \right]_0^1 = \frac{79\pi}{20}.
 \end{aligned}$$

■

Example 4 Find the volume of the solid generated by revolving the region described in Example 3 about the line $y = 3$.

Solution The region and a typical rectangle are resketched in (i) of Figure 6.19, together with the axis of revolution $y = 3$. The washer generated by the rectangle is illustrated in (ii) of Figure 6.19. Note that the radii of this washer are as follows:

$$\text{Inner radius} = 3 - (w_i^2 + 2) = 1 - w_i^2$$

$$\text{Outer radius} = 3 - \left(\frac{1}{2}w_i + 1 \right) = 2 - \frac{1}{2}w_i.$$

Using (6.5), the volume of the washer is

$$\pi \left[\left(2 - \frac{1}{2}w_i \right)^2 - (1 - w_i^2)^2 \right] \Delta x_i.$$

Taking a limit of a sum of such terms gives us

$$\begin{aligned}
 V &= \int_0^1 \pi \left[\left(2 - \frac{1}{2}x \right)^2 - (1 - x^2)^2 \right] dx \\
 &= \pi \int_0^1 \left[\left(4 - 2x + \frac{1}{4}x^2 \right) - (1 - 2x^2 + x^4) \right] dx \\
 &= \pi \int_0^1 \left(3 - 2x + \frac{9}{4}x^2 - x^4 \right) dx \\
 &= \pi \left[3x - x^2 + \frac{3}{4}x^3 - \frac{1}{5}x^5 \right]_0^1 \\
 &= \pi \left[3 - 1 + \frac{3}{4} - \frac{1}{5} \right] = \frac{51}{20}\pi \approx 8.01.
 \end{aligned}$$

■

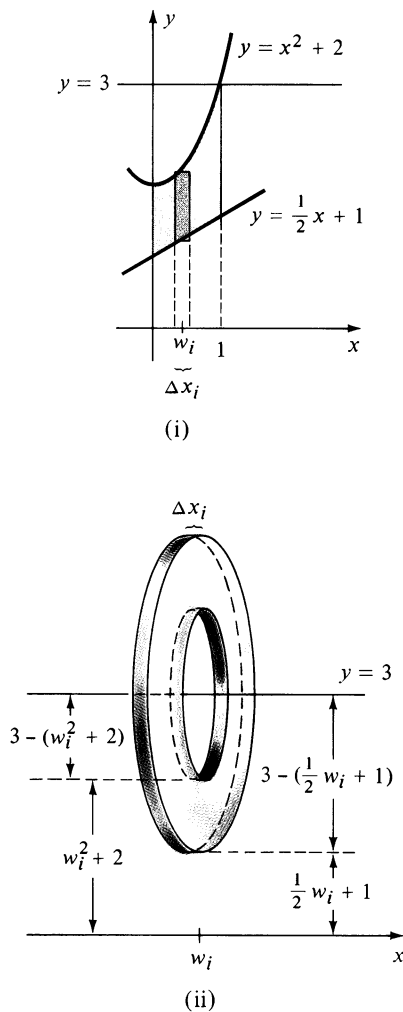


FIGURE 6.19

By interchanging the roles of x and y we can apply the techniques discussed in this section to solids generated by revolving regions about the y -axis, as in the next example.

Example 5 The region in the first quadrant bounded by the graphs of $y = \frac{1}{8}x^3$ and $y = 2x$ is revolved about the y -axis. Find the volume of the resulting solid.

Solution The region and a typical rectangle are shown in (i) of Figure 6.20. Since we wish to integrate with respect to y , we solve the given equations for x in terms of y , obtaining

$$x = \frac{1}{2}y \quad \text{and} \quad x = 2y^{1/3}.$$

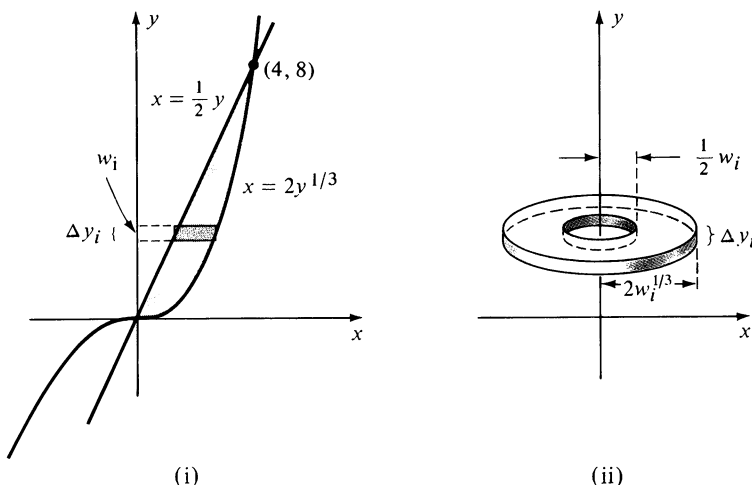


FIGURE 6.20

As shown in (ii) of Figure 6.20, the inner and outer radii of the washer generated by the rectangle are $\frac{1}{2}w_i$ and $2w_i^{1/3}$, respectively. Since the thickness is Δy_i it follows from (6.5) that the volume of the washer is

$$\pi \left[(2w_i^{1/3})^2 - \left(\frac{1}{2} w_i \right)^2 \right] \Delta y_i = \pi \left[4w_i^{2/3} - \frac{1}{4} w_i^2 \right] \Delta y_i.$$

Taking a limit of a sum of such terms gives us

$$\begin{aligned} V &= \int_0^8 \pi \left[4y^{2/3} - \frac{1}{4} y^2 \right] dy = \pi \left[\frac{12}{5} y^{5/3} - \frac{1}{12} y^3 \right]_0^8 \\ &= \pi \left[\frac{12}{5} (8^{5/3}) - \frac{1}{12} (8^3) \right] = \frac{512}{15} \pi \approx 107.2. \end{aligned}$$

6.2 Exercises

In Exercises 1–12 sketch the region R bounded by the graphs of the given equations and find the volume of the solid generated by revolving R about the indicated axis. In each case show a typical rectangle, together with the disc or washer it generates. (Remark: Although the solids described in the following exercises are mathematical entities, it should be kept in mind

that many objects encountered in everyday life have similar shapes.)

1 $y = 1/x$, $x = 1$, $x = 3$, $y = 0$; about the x -axis.

2 $y = \sqrt{x}$, $y = 0$, $x = 4$; about the x -axis.

- 3 $y = x^2$, $y = 2$; about the y -axis.
- 4 $y = 1/x$, $x = 0$, $y = 1$, $y = 3$; about the y -axis.
- 5 $y = x^2 - 4x$, $y = 0$; about the x -axis.
- 6 $y = x^3$, $x = -2$, $y = 0$; about the x -axis.
- 7 $y^2 = x$, $2y = x$; about the y -axis.
- 8 $y = 2x$, $y = 4x^2$; about the y -axis.
- 9 $y = x^2$, $y = 4 - x^2$; about the x -axis.
- 10 $x = y^3$, $x^2 + y = 0$; about the x -axis.
- 11 $x = y^2$, $y - x + 2 = 0$; about the y -axis.
- 12 $x + y = 1$, $y = x + 1$, $x = 2$; about the y -axis.
- 13 The region bounded by the x -axis and the graph of $y = \sin 2x$ from $x = 0$ to $x = \pi$ is revolved about the x -axis. Find the volume of the resulting solid. (*Hint*: $\sin^2 \theta = \frac{1}{2}(1 - \cos 2\theta)$.)
- 14 The region bounded by the x -axis and the graph of $y = 1 + \cos 3x$ from $x = 0$ to $x = 2\pi$ is revolved about the x -axis. Find the volume of the resulting solid. (*Hint*: $\cos^2 \theta = \frac{1}{2}(1 + \cos 2\theta)$.)
- 15 The region between the graphs of $y = \sin x$ and $y = \cos x$ from $x = 0$ to $x = \pi/4$ is revolved about the x -axis. Find the volume of the resulting solid. (*Hint*: Use a trigonometric identity to change the form of the integrand.)
- 16 The region bounded by the graphs of $y = \cos \frac{1}{2}x$, $y = 2$, $x = 0$ and $x = 4\pi$ is revolved about the line $y = 2$. Find the volume of the resulting solid.
- 17 Find the volume of the solid generated by revolving the region bounded by the graphs of $y = x^2$ and $y = 4$ (a) about the line $y = 4$; (b) about the line $y = 5$; (c) about the line $x = 2$.
- 18 Find the volume of the solid generated by revolving the region bounded by the graphs of $y = \sqrt{x}$, $y = 0$, and $x = 4$ (a) about the line $x = 4$; (b) about the line $x = 6$; (c) about the line $y = 2$.

In Exercises 19–24 sketch the region R bounded by the graphs of the equations given in (a) and then set up (but do not evaluate) integrals needed to find the volume of the solid obtained by revolving R about the line given in (b). As usual, show typical rectangles and the corresponding discs or washers.

- 19 (a) $y = x^3$, $y = 4x$
(b) $y = 8$
- 20 (a) $y = x^3$, $y = 4x$
(b) $x = 4$
- 21 (a) $x + y = 3$, $y + x^2 = 3$
(b) $x = 2$
- 22 (a) $y = 1 - x^2$, $x - y = 1$
(b) $y = 3$
- 23 (a) $x^2 + y^2 = 1$
(b) $x = 5$
- 24 (a) $y = x^{2/3}$, $y = x^2$
(b) $y = -1$

In Exercises 25–28 use a definite integral to derive a formula for the volume of the indicated solid.

- 25 A right circular cone of altitude h and radius of base r .
- 26 A sphere of radius r .
- 27 A frustum of a right circular cone of altitude h , lower base radius r_1 , and upper base radius r_2 .
- 28 The volume of a spherical segment of height h and radius of sphere r .

Exercises 29 and 30 each represent a limit of a sum for a function f on the interval $[0, 1]$. Interpret each limit as a volume and find its value.

- 29 $\lim_{\|P\| \rightarrow 0} \sum_i \pi(w_i^4 - w_i^6) \Delta x_i$
- 30 $\lim_{\|P\| \rightarrow 0} \sum_i \pi(w_i - w_i^8) \Delta y_i$

6.3 Volumes Using Cylindrical Shells

There is another method for finding volumes of solids of revolution which, in certain cases, is simpler to apply than those discussed in Section 6.2. The method employs hollow circular cylinders, that is, thin cylindrical shells of the type illustrated in Figure 6.21.

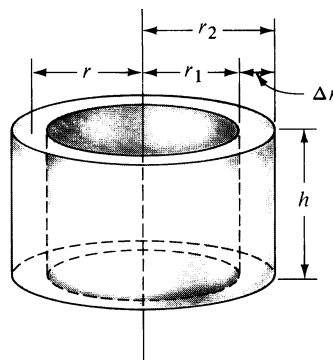


FIGURE 6.21

The volume of a shell having outer radius r_2 , inner radius r_1 , and altitude h is $\pi r_2^2 h - \pi r_1^2 h$. This expression may also be written

$$\begin{aligned}\pi(r_2^2 - r_1^2)h &= \pi(r_2 + r_1)(r_2 - r_1)h \\ &= 2\pi\left(\frac{r_2 + r_1}{2}\right)h(r_2 - r_1).\end{aligned}$$

If we let $r = (r_2 + r_1)/2$ (the **average radius** of the shell) and $\Delta r = r_2 - r_1$ (the **thickness** of the shell), then the volume of the shell is given by $2\pi r h \Delta r$, that is,

$$(6.6) \quad \text{Volume of a shell} = 2\pi(\text{average radius})(\text{altitude})(\text{thickness}).$$

Let the function f be continuous and $f(x) \geq 0$ for all x in $[a, b]$, where $0 \leq a < b$. Let R be the region bounded by the graph of f , the x -axis, and by the graphs of $x = a$ and $x = b$, as illustrated in (i) of Figure 6.22. The solid generated by revolving R about the y -axis is illustrated in (ii) of the figure. Note that if $a > 0$, there is a hole through the solid. Let P be a partition of $[a, b]$ and consider a rectangle with base corresponding to the interval $[x_{i-1}, x_i]$ and altitude $f(w_i)$, where w_i is the midpoint of $[x_{i-1}, x_i]$. If this rectangle is revolved about the y -axis, then as illustrated in (iii) of Figure 6.22 there results a cylindrical shell with average radius w_i , altitude $f(w_i)$, and thickness $\Delta x_i = x_i - x_{i-1}$. By (6.6) we may express its volume as

$$2\pi w_i f(w_i) \Delta x_i.$$

Doing this for each subinterval in the partition and adding gives us

$$\sum_i 2\pi w_i f(w_i) \Delta x_i.$$

Geometrically, this sum represents the volume of a solid of the type illustrated in (iv) of Figure 6.22. Evidently, the smaller the norm $\|P\|$ of the partition, the better the sum approximates the volume V of the solid generated by R .

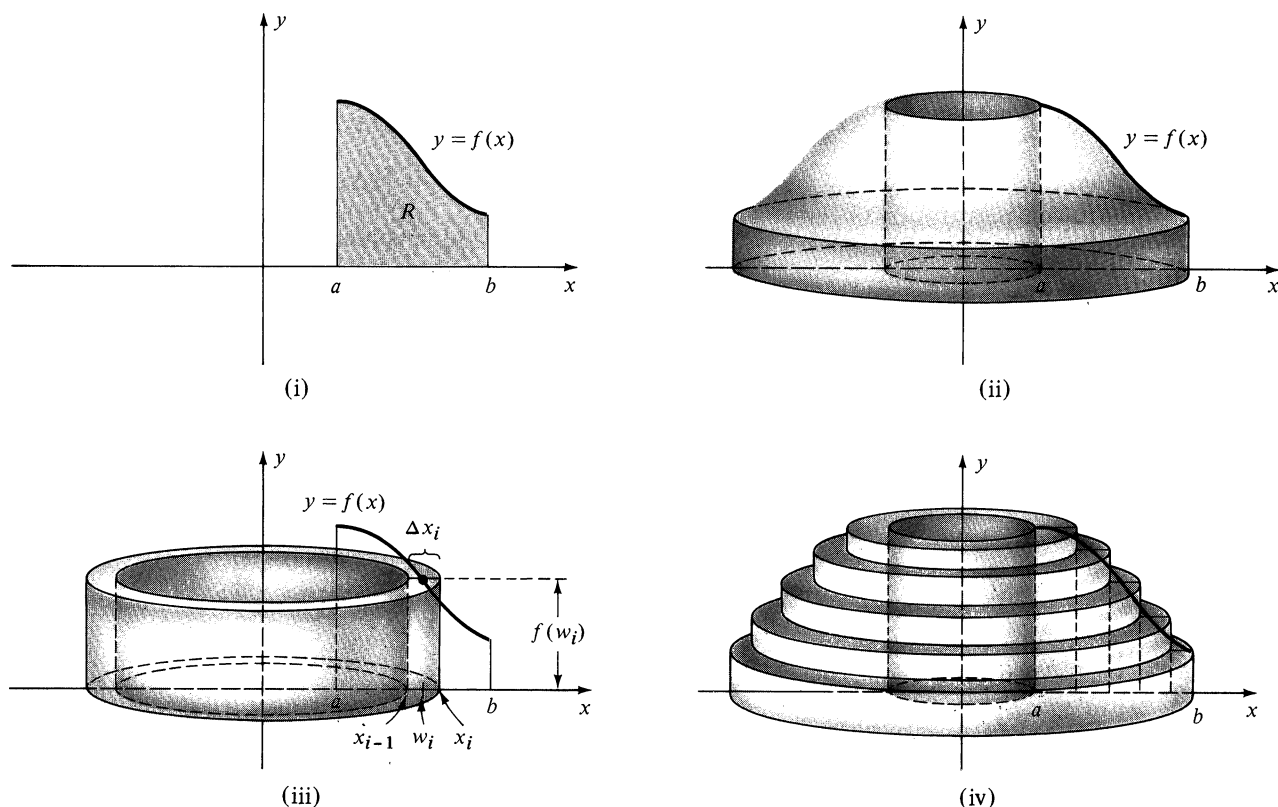


FIGURE 6.22

Indeed, it appears that the volume of the solid illustrated in (ii) of the figure is the limit of the sum. This gives us the following.

Definition (6.7)

Let f be continuous on $[a, b]$ where $0 \leq a < b$. The **volume** V of the solid of revolution generated by revolving the region bounded by the graphs of f , $x = a$, $x = b$, and the x -axis about the y -axis is

$$V = \lim_{\|P\| \rightarrow 0} \sum_i 2\pi w_i f(w_i) \Delta x_i = \int_a^b 2\pi x f(x) dx.$$

The last equality follows from the definition of the definite integral. It can be proved that if the methods of Section 6.2 are also applicable, then both methods lead to the same answer.

Example 1 The region bounded by the graph of $y = 2x - x^2$ and the x -axis is revolved about the y -axis. Find the volume of the resulting solid.

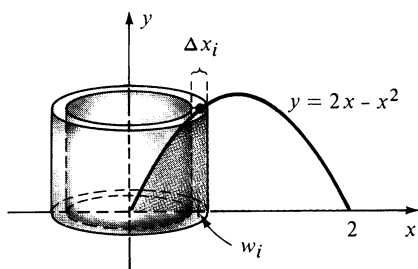


FIGURE 6.23

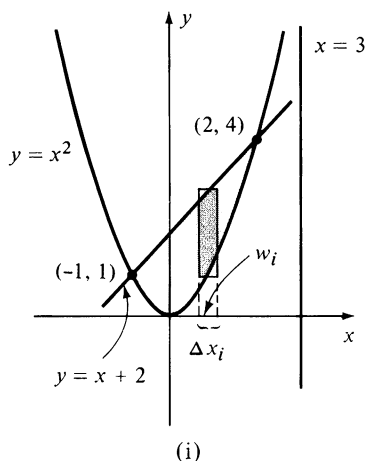
Solution The region and a shell generated by a typical rectangle are sketched in Figure 6.23, where w_i is the midpoint of $[x_{i-1}, x_i]$. Since the average radius of the shell is w_i , the altitude is $2w_i - w_i^2$, and the thickness is Δx_i , it follows from (6.6) that the volume of the shell is

$$2\pi w_i(2w_i - w_i^2) \Delta x_i.$$

Consequently, as in Definition (6.7), the volume V of the solid is given by

$$\begin{aligned} V &= \lim_{\|P\| \rightarrow 0} \sum_i 2\pi w_i(2w_i - w_i^2) \Delta x_i \\ &= \int_0^2 2\pi x(2x - x^2) dx = 2\pi \int_0^2 (2x^2 - x^3) dx \\ &= 2\pi \left[\frac{2}{3} x^3 - \frac{1}{4} x^4 \right]_0^2 = \frac{8\pi}{3}. \end{aligned}$$

The volume V can also be found using washers; however, the calculations would be more involved since the given equation would have to be solved for x in terms of y . ■



(i)

Example 2 The region bounded by the graphs of $y = x^2$ and $y = x + 2$ is revolved about the line $x = 3$. Express the volume of the resulting solid as a definite integral.

Solution The region and a typical rectangle are shown in (i) of Figure 6.24, where w_i represents the midpoint of the i th subinterval $[x_{i-1}, x_i]$. The cylindrical shell generated by the rectangle is illustrated in (ii) of Figure 6.24. This shell has the following dimensions:

$$\text{altitude} = (w_i + 2) - w_i^2$$

$$\text{average radius} = 3 - w_i$$

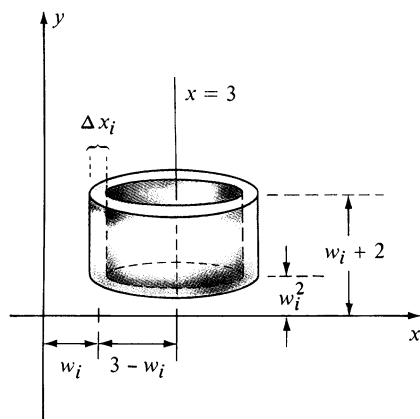
$$\text{thickness} = \Delta x_i.$$

Hence, by (6.6) its volume is

$$2\pi(3 - w_i)[(w_i + 2) - w_i^2] \Delta x_i.$$

Taking the limit of a sum of such terms gives us

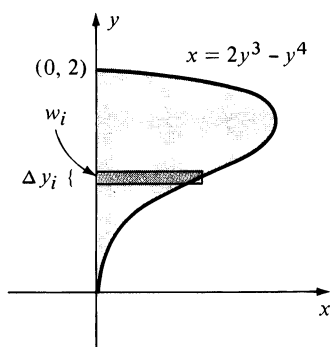
$$V = \int_{-1}^2 2\pi(3 - x)(x + 2 - x^2) dx. \quad \blacksquare$$



(ii)

FIGURE 6.24

The definite integral in the preceding example could be evaluated by multiplying the factors in the integrand and then integrating each term. Since we have worked a sufficient number of problems of this type, it would not be very instructive to carry out all of these details. For convenience we shall refer to the process of expressing V in terms of an integral as *setting up the integral for V* .



(i)

By interchanging the roles of x and y we can find certain volumes by using shells and integrating with respect to y , as illustrated in the next example.

Example 3 The region in the first quadrant bounded by the graph of $x = 2y^3 - y^4$ and the y -axis is revolved about the x -axis. Set up the integral for the volume of the resulting solid.

Solution The region is sketched in (i) of Figure 6.25 together with a typical (horizontal) rectangle. The cylindrical shell generated by the rectangle is illustrated in (ii) of Figure 6.25. The shell in the figure has the following dimensions:

$$\text{altitude} = 2w_i^3 - w_i^4$$

$$\text{average radius} = w_i$$

$$\text{thickness} = \Delta y_i.$$

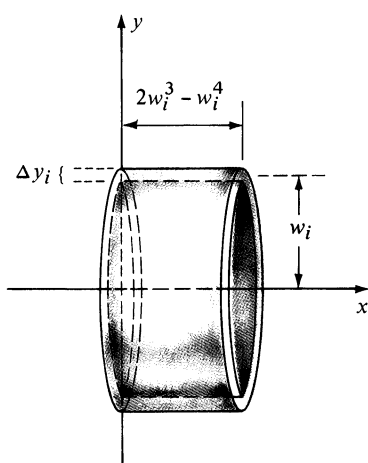
Hence, by (6.6) its volume is

$$2\pi w_i(2w_i^3 - w_i^4) \Delta y_i.$$

Taking a limit of a sum of such terms and remembering that y is the independent variable gives us

$$V = \int_0^2 2\pi y(2y^3 - y^4) dy.$$

■



(ii)

FIGURE 6.25

It is worth noting that in the preceding example we were forced to use shells and to integrate with respect to y , since use of washers and integration with respect to x would require that the equation $x = 2y^3 - y^4$ be solved for y in terms of x , a rather formidable task.

6.3 Exercises

In Exercises 1–10 sketch the region R bounded by the graphs of the given equations and use the methods of this section to find the volume of the solid generated by revolving R about the indicated axis. In each case show a typical rectangle together with the cylindrical shell it generates.

- 1 $y = \sqrt{x}$, $x = 4$, $y = 0$; about the y -axis.
- 2 $y = 1/x$, $x = 1$, $x = 2$, $y = 0$; about the y -axis.
- 3 $y = x^2$, $y^2 = 8x$; about the y -axis.
- 4 $y = x^2 - 5x$, $y = 0$; about the y -axis.
- 5 $2x - y - 12 = 0$, $x - 2y - 3 = 0$, $x = 4$; about the y -axis.

- 6 $y = x^3 + 1$, $x + 2y = 2$, $x = 1$; about the y -axis.
- 7 $x^2 = 4y$, $y = 4$; about the x -axis.
- 8 $y^3 = x$, $y = 3$, $x = 0$; about the x -axis.
- 9 $y = 2x$, $y = 6$, $x = 0$; about the x -axis.
- 10 $2y = x$, $y = 4$, $x = 1$; about the x -axis.

- 11 The region bounded by the x -axis and the graph of $y = \cos(x^2)$ from $x = 0$ to $x = \sqrt{\pi/2}$ is revolved about the y -axis. Find the volume of the resulting solid.

- 12 The region bounded by the x -axis and the graph of $y = x \sin(x^3)$ from $x = 0$ to $x = 1$ is revolved about the y -axis. Find the volume of the resulting solid.
- 13 Find the volume of the solid generated by revolving the region bounded by the graphs of $y = x^2 + 1$, $x = 0$, $x = 2$, and $y = 0$ (a) about the line $x = 3$; (b) about the line $x = -1$.
- 14 Find the volume of the solid generated by revolving the region bounded by the graphs of $y = 4 - x^2$ and $y = 0$ (a) about the line $x = 2$; (b) about the line $x = -3$.

15–26 Use the methods of this section to solve Exercises 17–28 of Section 6.2.

The expressions in Exercises 27 and 28 each represent a limit of a sum for a function f on the interval $[0, 1]$. Interpret each limit as a volume and find its value.

$$27 \quad \lim_{\|P\| \rightarrow 0} \sum_i 2\pi(w_i^2 - w_i^3) \Delta x_i$$

$$28 \quad \lim_{\|P\| \rightarrow 0} \sum_i 2\pi(w_i^5 + w_i^{3/2}) \Delta x_i$$

6.4 Volumes by Slicing

If a plane intersects a solid, then the region common to the plane and the solid is called a **cross-section** of the solid. In Section 6.2 we encountered circular and washer-shaped cross sections. We shall now consider solids that have the property that for every x in a closed interval $[a, b]$ on a coordinate line l , the plane perpendicular to l at the point with coordinate x intersects the solid in a cross section whose area is given by $A(x)$, where A is a continuous function on $[a, b]$. Figures 6.26 and 6.27 illustrate solids of the type we wish to discuss.

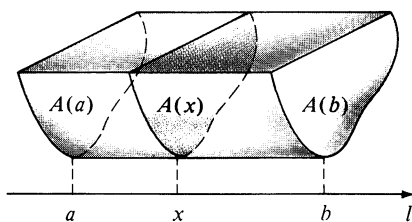


FIGURE 6.26

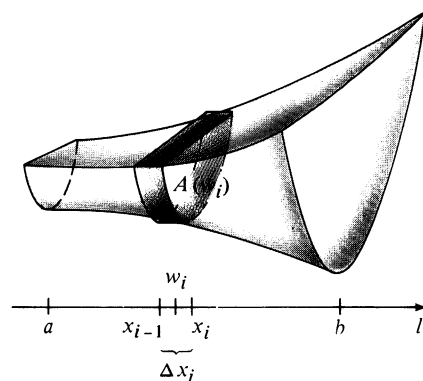


FIGURE 6.27

The solid is called a **cylinder** if, as illustrated in Figure 6.26, all cross sections are the same. If we are only interested in that part of the graph bounded by planes through the points with coordinates a and b , then the cross sections determined by these planes are called the **bases** of the cylinder and the distance between the bases is called the **altitude**. By definition, the volume of such a cylinder is the area of a base multiplied by the altitude. As a special case, a right circular cylinder of base radius r and altitude h has volume $\pi r^2 h$.

The solid illustrated in Figure 6.27 is not a cylinder since cross sections by planes perpendicular to l are not all the same. To find the volume we begin with a partition P of $[a, b]$ by choosing $a = x_0, x_1, x_2, \dots, x_n = b$. The planes perpendicular to l at the points with these coordinates slice the solid into smaller pieces. The i th such slice is shown in Figure 6.27. As usual, let $\Delta x_i = x_i - x_{i-1}$ and choose any number w_i in $[x_{i-1}, x_i]$. It appears that if Δx_i is small, then the volume of the slice can be approximated by the volume of the cylinder of base area $A(w_i)$ and altitude Δx_i , that is, by $A(w_i) \Delta x_i$. Consequently, the total volume of the solid is approximated by the Riemann sum $\sum_{i=1}^n A(w_i) \Delta x_i$. Since the approximation improves as $\|P\|$ gets smaller, we define the volume V of the solid as the limit of this sum. Our discussion may be summarized as follows, where it is assumed that the conditions on $A(x)$ are those we have imposed previously.

$$(6.8) \quad V = \lim_{\|P\| \rightarrow 0} \sum_i A(w_i) \Delta x_i = \int_a^b A(x) dx$$

where $A(x)$ is the area of a cross section corresponding to the number x in $[a, b]$.

Example 1 Find the volume of a right pyramid that has altitude h and square base of side a .

Solution If, as shown in Figure 6.28, we introduce a coordinate line l along the axis of the pyramid, with origin O at the vertex, then cross sections by planes perpendicular to l are squares. If $A(x)$ is the cross-sectional area determined by the plane that intersects the axis x units from O , then

$$A(x) = (2y)^2 = 4y^2$$

where y is the distance indicated in the figure. By similar triangles

$$\frac{y}{x} = \frac{a/2}{h}, \quad \text{or} \quad y = \frac{ax}{2h}$$

and hence

$$A(x) = 4y^2 = \frac{4a^2x^2}{4h^2} = \frac{a^2}{h^2} x^2.$$

Applying (6.8),

$$V = \int_0^h \left(\frac{a^2}{h^2} \right) x^2 dx = \left(\frac{a^2}{h^2} \right) \left[\frac{x^3}{3} \right]_0^h = \frac{a^2 h}{3}.$$

■

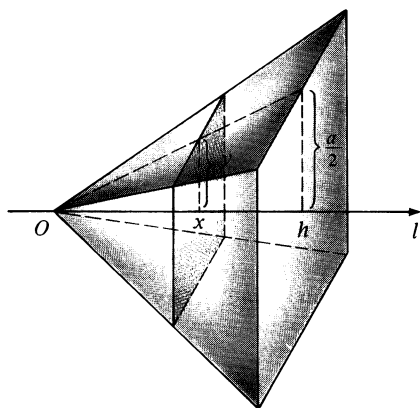


FIGURE 6.28

Example 2 A solid has, as its base, the circular region in the xy -plane bounded by the graph of $x^2 + y^2 = a^2$, where $a > 0$. Find the volume of the solid if every cross section by a plane perpendicular to the x -axis is an equilateral triangle with one side in the base.

Solution A typical cross section by a plane x units from the origin is illustrated in Figure 6.29.

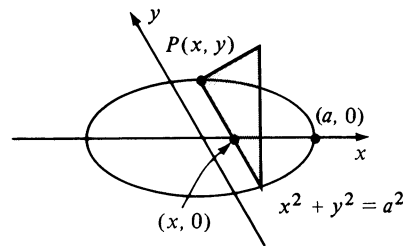


FIGURE 6.29

If the point $P(x, y)$ is on the circle, then the length of a side of the triangle is $2y$ and the altitude is $\sqrt{3}y$. Hence the area $A(x)$ of the pictured triangle is

$$A(x) = \frac{1}{2}(2y)(\sqrt{3}y) = \sqrt{3}y^2 = \sqrt{3}(a^2 - x^2).$$

Applying (6.8) gives us

$$V = \int_{-a}^a \sqrt{3}(a^2 - x^2) dx = \sqrt{3} \left[a^2x - \frac{x^3}{3} \right]_{-a}^a = \frac{4\sqrt{3}a^3}{3}. \quad \blacksquare$$

6.4 Exercises

- The base of a solid is the circular region in the xy -plane bounded by the graph of $x^2 + y^2 = a^2$, where $a > 0$. Find the volume of the solid if every cross section by a plane perpendicular to the x -axis is a square.
- Work Exercise 1 if every cross section is an isosceles triangle with base on the xy -plane and altitude equal to the length of the base.
- A solid has as its base the region in the xy -plane bounded by the graphs of $y = 4$ and $y = x^2$. Find the volume of the solid if every cross section by a plane perpendicular to the x -axis is an isosceles right triangle with hypotenuse on the xy -plane.
- Work Exercise 3 if each cross section is a square.
- Find the volume of a pyramid of the type illustrated in Figure 6.28 if the altitude is h and the base is a rectangle of dimensions a and $2a$.
- A solid has as its base the region in the xy -plane bounded by the graphs of $y = x$ and $y^2 = x$. Find the volume of the solid if every cross section by a plane perpendicular to the x -axis is a semicircle with diameter in the xy -plane.
- A solid has as its base the region in the xy -plane bounded by the graphs of $y^2 = 4x$ and $x = 4$. If every cross section by a plane perpendicular to the y -axis is a semicircle, find the volume of the solid.
- A solid has as its base the region in the xy -plane bounded by the graphs of $x^2 = 16y$ and $y = 2$. Every cross section by a plane perpendicular to the y -axis is a rectangle whose height is twice that of the side in the xy -plane. Find the volume of the solid.
- A log having the shape of a right circular cylinder of radius a is lying on its side. A wedge is removed from the log by making a vertical cut and another cut at an angle of 45° , both cuts intersecting at the center of the log (see figure). Find the volume of the wedge.

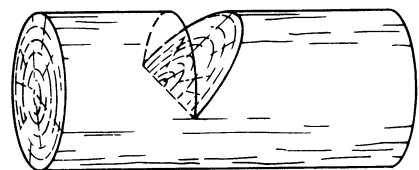


FIGURE FOR EXERCISE 9

- 10 The axes of two right circular cylinders of radius a intersect at right angles. Find the volume of the solid bounded by the cylinders. (Hint: Use square cross sections.)
- 11 The base of a solid is the circular region in the xy -plane bounded by the graph of $x^2 + y^2 = a^2$, where $a > 0$. Find the volume of the solid if every cross section by a plane perpendicular to the x -axis is an isosceles triangle of constant altitude b . (Hint: Interpret $\int_{-a}^a \sqrt{a^2 - x^2} dx$ as an area.)
- 12 Cross sections of a horn-shaped solid by planes perpendicular to its axis are circles. If a cross section which is s inches from the smaller end of the solid has diameter $6 + (s^2/36)$ in., and if the length of the solid is 2 ft, find its volume.
- 13 A tetrahedron has three mutually perpendicular faces and three mutually perpendicular edges of lengths 2, 3, and 4 cm, respectively. Find its volume.
- 14 A hole of diameter d is bored through a spherical solid of radius r such that the axis of the hole coincides with a diameter of the sphere. Find the volume of the solid that remains.
- 15 The base of a solid is an isosceles right triangle whose equal sides have length a . Find the volume if cross sections that are perpendicular to the base and to one of the equal sides are semi-circular.
- 16 Work Exercise 15 if the cross sections are regular hexagons with one side in the base.
- 17 A **prismatoid** is a solid whose cross-sectional areas by planes parallel to, and a distance x from, a fixed plane can be expressed as $ax^2 + bx + c$ where a , b , and c are real numbers. Prove that the volume V of a prismatoid is given by

$$V = \frac{h}{6}(B_1 + B_2 + 4B)$$

where B_1 and B_2 are the areas of the bases, B is the cross-sectional area parallel to and halfway between the bases, and h is the distance between the bases. Show that the frustum of a right circular cone of base radii r_1 and r_2 and altitude h is a prismatoid and find its volume.

- 18 *Cavalieri's Theorem* states that if two solids have equal altitudes and if all cross sections by planes parallel to their bases and at the same distances from their bases have equal areas, then the solids have the same volume. Prove this theorem.
- 19 Show that the disc and washer methods discussed in Section 6.2 are special cases of (6.8).
- 20 A circular swimming pool has diameter 28 ft. The depth of the water changes slowly from 3 ft at a point A on one side of the pool to 9 ft at a point B diametrically opposite A (see figure). Depth readings (in ft) taken along the diameter AB are given in the following table, where x is the distance (in ft) from A .

x	0	4	8	12	16	20	24	28
Water depth	3	3.5	4	5	6.5	8	8.5	9

Use the Trapezoidal Rule (with $n = 7$) to estimate the volume of water in the pool. Approximate the number of gallons of water contained in the pool (1 gal ≈ 0.134 ft³).

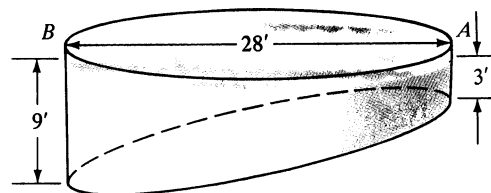


FIGURE FOR EXERCISE 20

6.5 Work

The concept of **force** may be thought of intuitively as the physical entity that is used to describe a push or pull on an object. For example, a force is needed to push or pull an object along a horizontal plane, to lift an object off the ground, or to move a charged particle through an electromagnetic field.

Forces are often measured in pounds. If an object weighs ten pounds, then by definition the force required to lift it (or hold it off the ground) is ten

pounds. A force of this type is a **constant force**, since its magnitude does not change while it is applied to the given object.

The concept of *work* is used when a force moves an object from one place to another. The following definition covers the simplest case, where the object moves along a line in the same direction as the applied force.

Definition (6.9)

If a constant force F is applied to an object, moving it a distance d in the direction of the force, then the **work** W done on the object is given by

$$W = Fd.$$

In the British system, if F is measured in pounds and d in feet, the unit for W is the foot-pound (ft-lb). If F is in pounds and d in inches, then the unit of work is the inch-pound (in.-lb). In the metric system, two different units of force are used, depending on whether the cgs (centimeter-gram-second) system or the mks (meter-kilogram-second) system is employed. In the cgs system, a **dyne** is defined as the force that, when applied to a mass of 1 gram, induces an acceleration of 1 cm/sec^2 . If F is expressed in dynes and d in centimeters, then the unit of work is the dyne-centimeter (dyn-cm), or **erg**. In the mks system, 1 **Newton** is the magnitude of the force required to impart an acceleration of 1 m/sec^2 to a mass of 1 kg. If F is given in Newtons and d in meters, then the unit of work is the Newton-meter (N-m), or **joule** (J). It can be shown that $1 \text{ joule} = 10^7 \text{ ergs} \approx 0.74 \text{ ft-lb}$.

Example 1 Find the work done in pushing an automobile along a level road from a point A to another point B , 20 feet from A , while exerting a constant force of 90 pounds.

Solution The problem is illustrated in Figure 6.30, where we have pictured the road as part of a line l . Since the constant force is $F = 90 \text{ lb}$ and the distance the automobile moves is $d = 20 \text{ ft}$, it follows from Definition (6.9) that the work done is

$$W = (90)(20) = 1,800 \text{ ft-lb}.$$

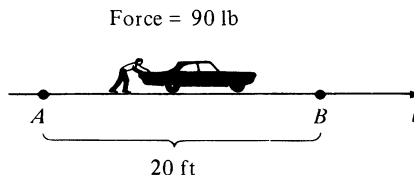


FIGURE 6.30

As illustrated in the next example, it is sometimes necessary to determine F before applying Definition (6.9). In this event the formula $F = ma$ from physics may be used, where m is the mass of the object and a is the acceleration.

Example 2 Approximate the work done if an object of mass 15 kg is lifted vertically, a distance of 4 meters.

Solution The required force F is given by $F = ma$, where if the mass m is measured in kg, and the acceleration a in m/sec^2 , then F is in Newtons. In this example, a is the acceleration due to gravity which is, in the mks system, approximately 9.81 m/sec^2 . Hence

$$F = ma \approx (15)(9.81) = 147.15 \text{ N}.$$

Applying Definition (6.9) and the mks system,

$$W = Fd \approx (147.15)(4) = 588.6 \text{ N}\cdot\text{m} = 588.6 \text{ joules.} \quad \blacksquare$$

Anyone who has pushed an automobile (or some other object) is aware of the fact that the force applied usually varies from one point to another. Thus, if an automobile is stalled it may require a larger force to get it moving than that which is needed after it is in motion. In addition, the force could vary considerably because of friction since, for example, part of the road may be smooth concrete and another part rough gravel. Similarly, someone inside the automobile could change the force required to move it by applying the brakes from time to time. Forces that are not constant are sometimes referred to as *variable forces*.

If a variable force is applied to an object, moving it a certain distance in the direction of the force, then methods of calculus are needed to find the work done. For the present we shall assume that the object moves along a line l . The case of motion along a nonlinear path will be considered in Chapter 18. Let us introduce a coordinate system on l and assume that the object moves from the point A with coordinate a to the point B with coordinate b where $b > a$. In order to solve the problem, it is essential to know the force at the point on l with coordinate x , for every x in the interval $[a, b]$. This force will be denoted by $f(x)$. For simplicity, it will be assumed that the function f obtained in this manner is continuous on $[a, b]$.

Let P denote the partition of the interval $[a, b]$ determined by the numbers $a = x_0, x_1, \dots, x_n = b$ and let $\Delta x_i = x_i - x_{i-1}$ (see Figure 6.31). If w_i is a number in $[x_{i-1}, x_i]$, then the force at the point Q with coordinate w_i is $f(w_i)$. If Δx_i is small, then since f is continuous, the values of f change very little in the interval $[x_{i-1}, x_i]$. Roughly speaking, the function f is almost constant on $[x_{i-1}, x_i]$. It appears, therefore, that the work W_i done as the object moves through the i th subinterval may be approximated by means of Definition (6.9); that is,

$$W_i \approx f(w_i) \Delta x_i.$$

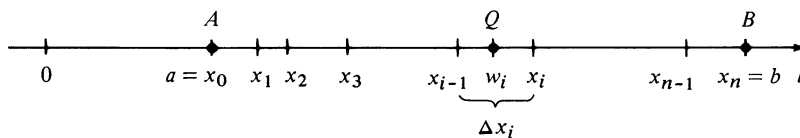


FIGURE 6.31

It seems evident that the smaller we choose Δx_i , the better $f(w_i) \Delta x_i$ approximates the work done in the interval $[x_{i-1}, x_i]$. If it is also assumed that work is additive, in the sense that the work W done as the object moves from A to B can be found by adding the work done over each subinterval, then

$$W \approx \sum_{i=1}^n f(w_i) \Delta x_i.$$

Since we expect this approximation to improve as the norm $\|P\|$ of the partition becomes smaller, it is natural to define W as the limit of the preceding sum. This limit leads to a definite integral.

Definition (6.10)

Let the force at the point with coordinate x on a coordinate line l be $f(x)$, where f is continuous on $[a, b]$. The **work** W done in moving an object from the point with coordinate a to the point with coordinate b is

$$W = \lim_{\|P\| \rightarrow 0} \sum_i f(w_i) \Delta x_i = \int_a^b f(x) dx.$$

The formula in Definition (6.10) can be used to find the work done in stretching or compressing a spring. To solve problems of this type it is necessary to use the following law from physics.

Hooke's Law (6.11)

The force $f(x)$ required to stretch a spring x units beyond its natural length is

$$f(x) = kx$$

where k is a constant called the **spring constant**.

The formula in (6.11) is also used to find the work done in compressing a spring x units from its natural length.

Example 3 A force of 9 lb is required to stretch a spring from its natural length of 6 in. to a length of 8 in. Find the work done in stretching the spring (a) from its natural length to a length of 10 in.; (b) from a length of 7 in. to a length of 9 in.

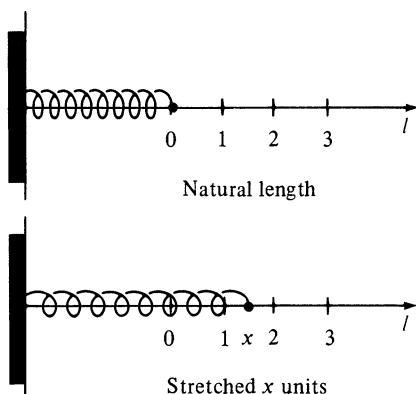


FIGURE 6.32

Solution

(a) Let us introduce a coordinate line l as shown in Figure 6.32, where one end of the spring is attached to some point to the left of the origin and the end to be pulled is located at the origin. According to Hooke's Law (6.11), the force $f(x)$ required to stretch a spring x units beyond its natural length is given by

$$f(x) = kx$$

for some constant k . Using the given data, $f(2) = 9$. Substituting in $f(x) = kx$ we obtain $9 = k \cdot 2$, and hence the spring constant is $k = \frac{9}{2}$. Consequently, for this spring, Hooke's Law has the form

$$f(x) = \frac{9}{2}x.$$

By Definition (6.10) the work done in stretching the spring 4 in. is given by

$$W = \int_0^4 \frac{9}{2}x \, dx = \frac{9}{4}x^2 \Big|_0^4 = 36 \text{ in.-lb.}$$

(b) We use the same function f but change the interval to $[1, 3]$, obtaining

$$W = \int_1^3 \frac{9}{2}x \, dx = \frac{9}{4}x^2 \Big|_1^3 = \frac{81}{4} - \frac{9}{4} = 18 \text{ in.-lb.} \quad \blacksquare$$

Example 4 A right circular conical tank of altitude 20 ft and radius of base 5 ft has its vertex at ground level and axis vertical. If the tank is full of water, find the work done in pumping the water over the top of the tank.

Solution We begin by introducing a coordinate system as shown in Figure 6.33. The cone intersects the xy -plane along the line of slope 4 through the origin. An equation for this line is $y = 4x$.

Let P denote the partition of the interval $[0, 20]$ determined by $0 = y_0, y_1, y_2, \dots, y_n = 20$, let $\Delta y_i = y_i - y_{i-1}$, and let x_i be the x -coordinate of the point on $y = 4x$ with y -coordinate y_i . If the cone is subdivided by means of planes perpendicular to the y -axis at each y_i , then we may think of the water as being sliced into n parts. As illustrated in Figure 6.33, the volume of the i th slice may be approximated by the volume $\pi x_i^2 \Delta y_i$ of a circular disc or, since $x_i = y_i/4$, by $\pi(y_i/4)^2 \Delta y_i$. This leads to the approximation

$$\text{Volume of } i\text{th slice} \approx \pi \left(\frac{y_i^2}{16} \right) \Delta y_i.$$

Assuming that water weighs 62.5 lb/ft^3 , the weight of the disc in Figure 6.33 is approximately $62.5\pi(y_i^2/16) \Delta y_i$. By (6.9), the work done in lifting the disc to the top of the tank is the product of the distance $20 - y_i$ and the weight, that is,

$$\text{Work done in lifting } i\text{th slice} \approx (20 - y_i)62.5\pi \left(\frac{y_i^2}{16} \right) \Delta y_i.$$

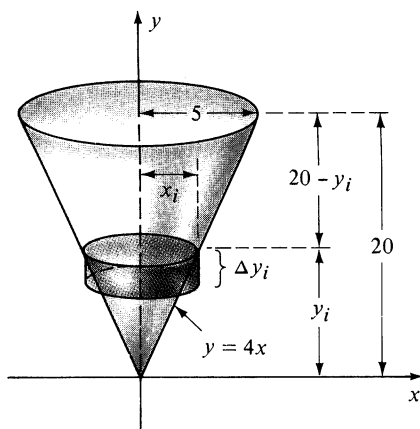


FIGURE 6.33

Since the last number is an approximation to the work done in lifting the i th slice to the top, the work done in emptying the entire tank is approximately

$$\sum_{i=1}^n (20 - y_i) 62.5\pi \left(\frac{y_i^2}{16} \right) \Delta y_i.$$

The actual work W is obtained by taking the limit of this sum as the norm $\|P\|$ approaches zero. This gives us

$$\begin{aligned} W &= \int_0^{20} (20 - y) 62.5\pi \left(\frac{y^2}{16} \right) dy \\ &= \frac{62.5\pi}{16} \int_0^{20} (20y^2 - y^3) dy \\ &= \frac{62.5\pi}{16} \left[\frac{20y^3}{3} - \frac{y^4}{4} \right]_0^{20} \\ &= \frac{62.5\pi}{16} \left(\frac{40,000}{3} \right) \approx 163,625 \text{ ft}\cdot\text{lb}. \end{aligned}$$

■

Example 5 The pressure p (lb/in.²) and volume v (in.³) of an enclosed expanding gas are related by the formula $pv^k = c$, where k and c are constants. If the gas expands from $v = a$ to $v = b$, show that the work done (in.-lb) is given by

$$W = \int_a^b p \, dv.$$

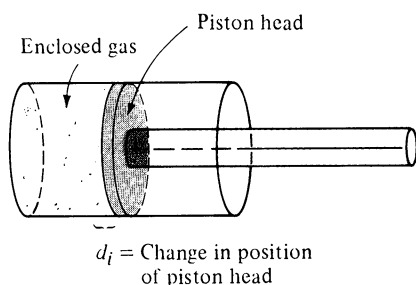


FIGURE 6.34

Solution Since the work done is independent of the shape of the container, we may assume that the gas is enclosed in a right circular cylinder of radius r and that the expansion takes place against a piston head as illustrated in Figure 6.34. Let P denote a partition of $[a, b]$ determined by $a = v_0, v_1, \dots, v_n = b$, and let $\Delta v_i = v_i - v_{i-1}$. We shall regard the expansion as taking place in volume increments $\Delta v_1, \Delta v_2, \dots, \Delta v_n$, and let $d_1, d_2, \dots, d_n$ be the corresponding distances that the piston head moves (see Figure 6.34). It follows that for each i ,

$$\Delta v_i = \pi r^2 d_i \quad \text{or} \quad d_i = \left(\frac{1}{\pi r^2} \right) \Delta v_i.$$

If p_i represents a value of the pressure p corresponding to the i th increment, then the force against the piston head is the product of p_i and the area of the piston head, that is, $p_i \pi r^2$. Hence the work done in this increment is

$$p_i \pi r^2 (d_i) = p_i \pi r^2 \left(\frac{1}{\pi r^2} \right) \Delta v_i = p_i \Delta v_i$$

and, therefore,

$$W \approx \sum_i p_i \Delta v_i.$$

Since this approximation improves as the Δv_i approach zero, we conclude that

$$W = \lim_{\|P\| \rightarrow 0} \sum_i p_i \Delta v_i = \int_a^b p \, dv. \quad \blacksquare$$

6.5 Exercises

- 1 A spring of natural length 10 in. stretches 1.5 in. under a weight of 8 lb. Find the work done in stretching the spring (a) from its natural length to a length of 14 in.; (b) from a length of 11 in. to a length of 13 in.
- 2 A force of 25 N is required to compress a spring of natural length 0.80 m to a length of 0.75 m. Find the work done in compressing the spring from its natural length to a length of 0.70 m.
- 3 If a spring is 12 cm long, compare the work done in stretching it from 12 cm to 13 cm with that done in stretching it from 13 cm to 14 cm.
- 4 It requires 60 ergs of work to stretch a certain spring from a length of 6 cm to 7 cm, and another 120 ergs of work to stretch it from 7 cm to 8 cm. Find the spring constant and the natural length of the spring.
- 5 A fishtank has a rectangular base of width 2 ft and length 4 ft, and rectangular sides of height 3 ft. If the tank is filled with water, what work is required to pump the water out over the top of the tank?
- 6 Generalize Example 4 of this section to the case of a conical tank of altitude h and radius of base a which is filled with a liquid of density (weight per unit volume) ρ .
- 7 A freight elevator of mass 1500 kg is supported by a cable 4 m long whose mass is 7 kg per linear meter. Approximate the work required to lift the elevator 3 m by winding the cable onto a winch.
- 8 A 170-pound man climbs a vertical telephone pole 15 ft high. What work is done if he reaches the top in (a) 10 sec? (b) 5 sec?
- 9 A vertical cylindrical tank of diameter 3 ft and height 6 ft is full of water. Find the work required to pump the water (a) out over the top of the tank; (b) out through a pipe which rises to a height 4 ft above the top of the tank.
- 10 Answer parts (a) and (b) of Exercise 9 if the tank is only half full of water.
- 11 The ends of a water trough 8 ft long are equilateral triangles having sides of width 2 ft. If the trough is full of water, find the work required to pump it out over the top.
- 12 A cistern has the shape of a hemisphere of radius 5 ft, with the circular part at the top. If the cistern is full of water, find the work required to pump all the water to a point 4 ft above the top of the cistern.
- 13 The force (in dyn) with which two electrons repel one another is inversely proportional to the square of the distance (in cm) between them.
 - (a) If one electron is held fixed at the point (5, 0), find the work done in moving a second electron along the x -axis from the origin to the point (3, 0).
 - (b) If two electrons are held fixed at the points (5, 0) and (−5, 0), respectively, find the work done in moving a third electron from the origin to (3, 0).
- 14 A uniform cable 12 m long and having mass 30 kg hangs vertically from a pulley system at the top of a building (see figure). If a steel beam of mass 250 kg is attached to the end of the cable, what work is required to pull it to the top?

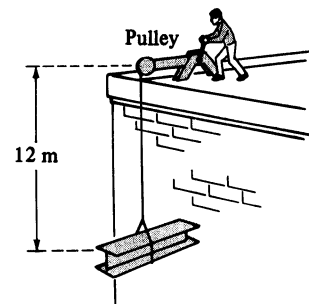


FIGURE FOR EXERCISE 14

- 15 A bucket containing water is lifted vertically at a constant rate of 1.5 ft/sec by means of a rope of negligible weight. As it rises, water leaks out at the rate of 0.25 lb/sec. If the bucket weighs 4 lb when empty, and if it contained 20 lb of water at the instant that the lifting began, determine the work done in raising the bucket 12 ft.
- 16 In Exercise 15, find the work required to raise the bucket until half the water has leaked out.

- 17 The volume and pressure of a certain gas vary in accordance with the law $pv^{1.2} = 115$, where the units of measurement are inches and pounds. Find the work done if the gas expands from 32 to 40 in.³. (Hint: See Example 5.)
- 18 The pressure and volume of a quantity of enclosed steam are related by the formula $pv^{1.14} = c$, where c is a constant. If the initial pressure and volume are p_0 and v_0 , respectively, find a formula for the work done if the steam expands to twice its volume. (Hint: See Example 5.)
- 19 Newton's Law of Gravitation states that the force F of attraction between two particles having masses m_1 and m_2 is given by $F = gm_1m_2/s^2$, where g is a constant and s is the distance between the particles. If the mass m_1 of the Earth is regarded as concentrated at the center of the Earth, and a rocket of mass m_2 is on the surface (a distance of 4,000 mi from the center), find a general formula for the work done in firing the rocket vertically upward to an altitude h (see figure).

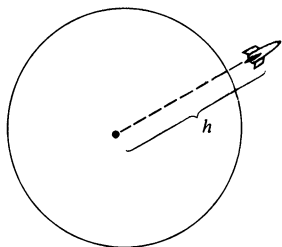


FIGURE FOR EXERCISE 19

- 20 In the study of electricity the formula $F = q/kr^2$, where k is a constant, is used to find the force (in dyn) with which a positive charge Q of strength q units repels a unit positive charge located r cm from Q . Find the work done in moving a unit charge from a point d cm from Q to a point $d/2$ cm from Q .

Suppose the tables in Exercises 21 and 22 were obtained experimentally, where $f(x)$ denotes the force acting at the point with coordinate x on a coordinate line l . Use the Trapezoidal Rule (5.35) to approximate the work done on the interval $[a, b]$, where a and b are the smallest and largest values of x , respectively.

21	x ft	$f(x)$ lb	22	x m	$f(x)$ N
	0	7.4		1	125
	0.5	8.1		2	120
	1.0	8.4		3	130
	1.5	7.8		4	146
	2.0	6.3		5	165
	2.5	7.1		6	157
	3.0	5.9		7	150
	3.5	6.8		8	143
	4.0	7.0		9	140
	4.5	8.0			
	5.0	9.2			

- 23 If the force function is constant, show that Definition (6.10) reduces to Definition (6.9).

6.6 Force Exerted by a Liquid

In physics the *pressure* p at a depth h in a liquid is defined as the weight of the liquid contained in a column having a cross-sectional area of one square unit and altitude h . Pressure may also be regarded as the force per unit area exerted by the liquid, as in the following definition.

Definition (6.12)

If the **density** (the weight per unit volume) of a liquid is denoted by ρ , then the **pressure** p at a depth h is

$$p = \rho h.$$

It can be verified experimentally that the pressure at any depth is exerted equally in all directions.

Example 1 If the density of the water in a lake is 62.5 lb/ft^3 , find the pressure at a depth of (a) 4 ft; (b) 10 ft.

Solution Using Definition (6.12) we obtain

$$(a) \quad p = \rho h = (62.5)(4) = 250 \text{ lb/ft}^2.$$

$$(b) \quad p = \rho h = (62.5)(10) = 625 \text{ lb/ft}^2. \quad \blacksquare$$

If a rectangular tank, such as an ordinary fish aquarium, is filled with water, then the total force exerted by the water on the (horizontal) base can be found by multiplying the pressure at the bottom of the tank by the area of the base. For example, if the depth of water is 4 ft and the area of the base is 10 ft^2 , then from (a) of Example 1, the pressure at the bottom is 250 lb/ft^2 and the total force acting on the base is $(250)(10) = 2500 \text{ lb}$. This corresponds to 10 columns of water, each having cross-sectional area 1 ft^2 and each weighing 250 lb.

It is more complicated to find the force exerted on one of the sides of the tank since the pressure is not constant there, but increases as the depth increases. Instead of investigating this particular problem, we shall consider a more general situation.

Consider a flat plate that is submerged in a liquid of density ρ such that the face of the plate is perpendicular to the surface of the liquid. Suppose that the shape of the plate is the same as that of the region in an xy -plane bounded by the graphs of the equations $y = c$, $y = d$, $x = f(y)$, and $x = g(y)$, where f and g are continuous functions on $[c, d]$ and $f(y) \geq g(y)$ for all y in $[c, d]$. Furthermore, suppose that the surface of the water contains the line $y = k$. A region of this type is illustrated in Figure 6.35, although in general it is not required that the region lie entirely above the x -axis as shown in the figure.

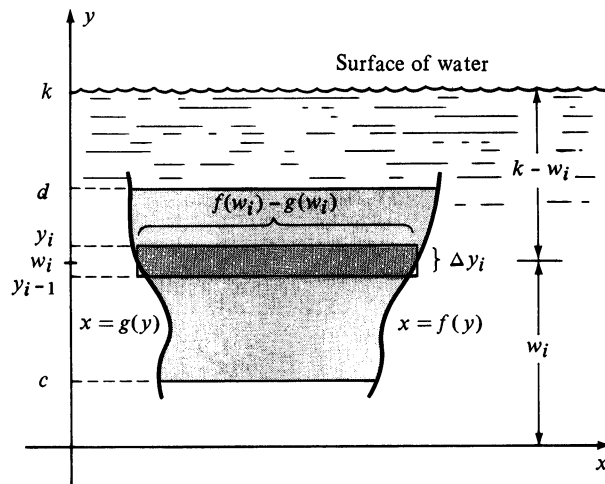


FIGURE 6.35

Let P denote the partition of $[c, d]$ determined by $c = y_0, y_1, \dots, y_n = d$ and let $\Delta y_i = y_i - y_{i-1}$. Select a number w_i in each subinterval $[y_{i-1}, y_i]$ and consider the rectangle of area $[f(w_i) - g(w_i)] \Delta y_i$. If Δy_i is small, then

all points in the rectangular region are roughly the same distance, $k - w_i$, from the surface of the liquid and hence by Definition (6.12) the pressure at any point within the rectangle may be approximated by $\rho(k - w_i)$. It appears, therefore, that the force on the i th rectangle is approximately equal to this pressure times the area of the rectangle, that is,

$$\rho(k - w_i)[f(w_i) - g(w_i)] \Delta y_i.$$

The total force on the region shown in Figure 6.35 may be approximated by the Riemann sum

$$\sum_i \rho(k - w_i)[f(w_i) - g(w_i)] \Delta y_i.$$

Since we expect this approximation to improve as the norm of the partition decreases we arrive at the following definition.

Definition (6.13)

The force F exerted by a liquid of constant density ρ on a region of the type illustrated in Figure 6.35, where the functions f and g are continuous on $[c, d]$, is

$$\begin{aligned} F &= \lim_{\|P\| \rightarrow 0} \sum_i \rho(k - w_i)[f(w_i) - g(w_i)] \Delta y_i \\ &= \int_c^d \rho(k - y)[f(y) - g(y)] dy. \end{aligned}$$

If a more complicated region is divided into subregions of the type used in (6.13), we apply the definition to each subregion and add the resulting numbers. The coordinate system may be introduced in various ways. In Example 3 we shall choose the x -axis along the surface of the liquid and the positive direction of the y -axis downward. In this case, Definition (6.13) must be changed accordingly.

It is often convenient to use the following formula to approximate the force on a thin horizontal rectangle:

$$(6.14) \quad \text{Force} \approx (\text{density}) \cdot (\text{depth}) \cdot (\text{area of rectangle})$$

After using (6.14), we take a limit of a sum to obtain the total force on the region, as illustrated in the following examples.

Example 2 The ends of a water trough 8 ft long have the shape of isosceles trapezoids of lower base 4 ft, upper base 6 ft, and altitude 4 ft. Find the total force on one end if the trough is full of water.

Solution Figure 6.36 illustrates one of the ends of the trough superimposed on a rectangular coordinate system and appropriately labeled. As indicated in the figure, the equation of the line through the points (2, 0) and (3, 4) is $y = 4x - 8$ or, equivalently, $x = \frac{1}{4}(y + 8)$. It follows that the rectangle has area

$$2x_i \Delta y_i = 2\left(\frac{1}{4}\right)(w_i + 8) \Delta y_i = \left(\frac{1}{2}\right)(w_i + 8) \Delta y_i.$$

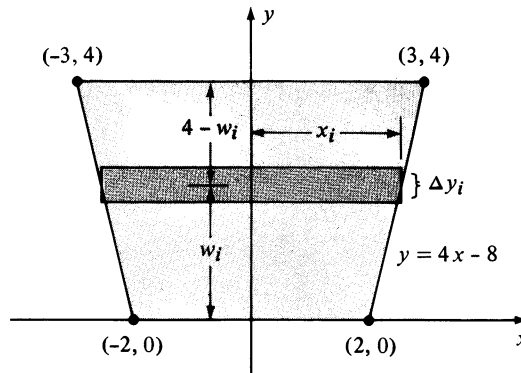


FIGURE 6.36

Using (6.14), we find that the force exerted by the liquid on this rectangle is approximately

$$(62.5)(4 - w_i)\left(\frac{1}{2}\right)(w_i + 8)\Delta y_i.$$

Summing and taking the limit gives us the total force. Thus,

$$\begin{aligned} F &= \int_0^4 (62.5)(4 - y)\left(\frac{1}{2}\right)(y + 8) dy \\ &= 31.25 \int_0^4 (32 - 4y - y^2) dy \\ &= 31.25 \left[32y - 2y^2 - \frac{y^3}{3} \right]_0^4 \\ &= 31.25 \left[128 - 32 - \frac{64}{3} \right] = \frac{7,000}{3} \text{ lb.} \end{aligned}$$

Example 3 A cylindrical oil storage tank 6 ft in diameter and 10 ft long is lying on its side. If the tank is half full of oil weighing 58 lb/ft³, find the force exerted by the oil on one side of the tank.

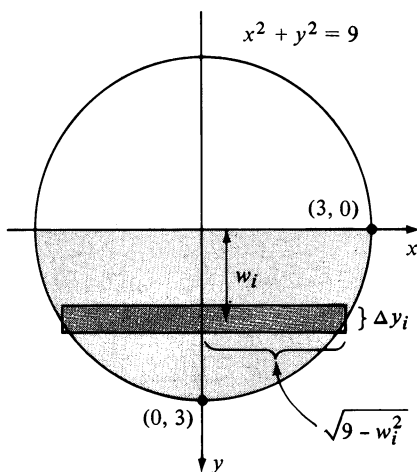


FIGURE 6.37

Solution Let us introduce a coordinate system so that the end of the tank is a circle of radius 3 ft with center at the origin. The equation of the circle is $x^2 + y^2 = 9$. If we choose the positive direction of the y -axis *downward*, then as shown in Figure 6.37, w_i represents the depth at a point in a typical horizontal rectangle. From the equation of the circle, the length of the rectangle is $2\sqrt{9 - w_i^2}$. Using (6.14), the force acting on the pictured rectangle is

$$58w_i(2\sqrt{9 - w_i^2})\Delta y_i.$$

As usual, we take a limit of a sum of such terms to obtain the total force F on the end of the tank. This gives us

$$F = \int_0^3 58y(2\sqrt{9 - y^2}) dy.$$

The integral may be evaluated by letting

$$u = 9 - y^2 \quad \text{and} \quad du = -2y dy.$$

Note that if $y = 0$, then $u = 9$, whereas if $y = 3$, then $u = 0$. We may, therefore, transform the integral as follows:

$$\begin{aligned} F &= -58 \int_0^3 \sqrt{9 - y^2} (-2y \, dy) \\ &= -58 \int_9^0 u^{1/2} \, du = -58 \left[\frac{u^{3/2}}{(3/2)} \right]_9^0 \\ &= -\frac{116}{3} \left[u^{3/2} \right]_9^0 = -\frac{116}{3} [0 - 9^{3/2}] = 1044 \text{ lb.} \quad \blacksquare \end{aligned}$$

6.6 Exercises

- 1 A glass tank to be used as an aquarium is 3 ft long and has square ends of width 1 ft. If the tank is filled with water, find the force exerted by the water on (a) one end and (b) one side.
- 2 If one of the square ends of the tank in Exercise 1 is divided into two parts by means of a diagonal, find the force exerted on each part.
- 3 The ends of a water trough 6 ft long have the shape of isosceles triangles with equal sides of length 2 ft and the third side of length $2\sqrt{3}$ ft at the top of the trough. Find the force exerted by the water on one end of the trough if the trough is (a) full of water; (b) half full of water.
- 4 Answer (a) and (b) of Exercise 3 if the (vertical) altitude of the triangle is h feet, where $0 < h < 2$.
- 5 A cylindrical oil storage tank 4 ft in diameter and 5 ft long is lying on its side. If the tank is half full of oil weighing 60 lb/ft^3 , find the force exerted by the oil on one end of the tank.
- 6 A rectangular gate in a dam is 5 ft long and 3 ft high. If the gate is vertical, with the top of the gate parallel to the surface of the water and 6 ft below it, find the force of the water against the gate.
- 7 A rectangular swimming pool is 20 ft wide and 40 ft long. The depth of the water in the pool varies uniformly from 3 ft at one end to 9 ft at the other end. Find the total force exerted by the water on the bottom of the pool.
- 8 Find the force exerted by the water on the side of the swimming pool described in Exercise 7.
- 9 A plate having the shape of an isosceles trapezoid with upper base 4 ft long and lower base 8 ft long is submerged vertically in water such that the bases are parallel to the surface. If the distances from the surface of the water to the lower and upper bases are 10 ft and 6 ft, respectively, find the force exerted by the water on one side of the plate.
- 10 A circular plate of radius 2 ft is submerged vertically in water. If the distance from the surface of the water to the center of the plate is 6 ft, find the force exerted by the water on one side of the plate.
- 11 The ends of a water trough have the shape of the region bounded by the graphs of $y = x^2$ and $y = 4$ where x and y are measured in feet. If the trough is full of water, find the force on one end.
- 12 A flat plate has the shape of the region bounded by the graphs of $y = x^4$ and $y = 1$ where x and y are measured in feet. The plate is submerged vertically in water with the straight part of its boundary parallel to (and closest to) the surface. If the distance from the surface of the water to the straight part of the boundary is 4 ft, find the force exerted by the water on one side of the plate.
- 13 A rectangular plate 3 ft wide and 6 ft long is submerged vertically in oil with its short side parallel to, and 2 ft below, the surface. If the oil weighs 50 lb/ft^3 , find the total force exerted on one side of the plate.
- 14 If the plate in Exercise 13 is divided into two parts by means of a diagonal, find the force exerted on each part.
- 15 A flat, irregularly-shaped plate is submerged vertically in water (see figure). Measurements of its width, taken at

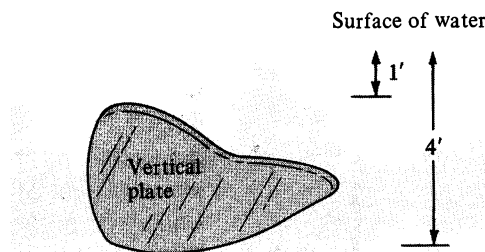


FIGURE FOR EXERCISE 15

successive depths of 0.5 ft, are compiled in the following table.

Water depth (ft)	1	1.5	2	2.5	3	3.5	4
Width of plate (ft)	0	2	3	5.5	4.5	3.5	0

Estimate the force of the water on one side of the plate by using (with $n = 6$) (a) the Trapezoidal Rule; (b) Simpson's Rule.

6.7 Arc Length

To solve certain problems in the sciences it is essential to consider the *length* of the graph of a function. For example, if a projectile moves along a parabolic course, we may wish to determine the distance it travels during a specified interval of time. Similarly, it may be necessary to find the length of a twisted piece of wire. Of course, if the wire is flexible, we could simply straighten it and find the linear length with a ruler (or by means of the Distance Formula). However, if the wire is not flexible, then other methods must be employed. As we shall see, the key to defining the length of a graph is to divide the graph into many small pieces and then approximate each piece by means of a line segment. Next, we take a limit of a sum of the lengths of all such line segments. This leads to a definite integral. To guarantee that the integral exists, it is necessary to place restrictions on the function, as indicated in the following discussion.

A function f is said to be **smooth** on an interval if it has a derivative f' that is continuous throughout the interval. Roughly speaking, this means that a small change in x produces a small change in the slope $f'(x)$ of the tangent line to the graph of f . Thus, there are no sharp corners on the graph. We intend to define what is meant by the **length of arc** between the two points A and B on the graph of a smooth function.

If f is smooth on a closed interval $[a, b]$, the points $A(a, f(a))$ and $B(b, f(b))$ will be called the **endpoints** of the graph of f . Let us consider the partition P of $[a, b]$ determined by $a = x_0, x_1, x_2, \dots, x_n = b$, and let Q_i denote the point with coordinates $(x_i, f(x_i))$. This gives us $n + 1$ points $Q_0, Q_1, Q_2, \dots, Q_n$ on the graph of f . If, as illustrated in Figure 6.38, we connect each Q_{i-1} to Q_i by a line segment of length $d(Q_{i-1}, Q_i)$, then the length L_P of the resulting broken line is

$$L_P = \sum_{i=1}^n d(Q_{i-1}, Q_i).$$

If the norm $\|P\|$ of the partition is small, then Q_{i-1} is close to Q_i for each i , and we expect L_P to be an approximation to the length of arc between A and B . This gives us a clue to a suitable definition of arc length. Specifically, we shall consider the limit of the sum L_P as $\|P\| \rightarrow 0$. To formulate this concept precisely, and at the same time arrive at a formula for calculating arc length, let us proceed as follows. By the Distance Formula,

$$d(Q_{i-1}, Q_i) = \sqrt{(x_i - x_{i-1})^2 + [f(x_i) - f(x_{i-1})]^2}.$$

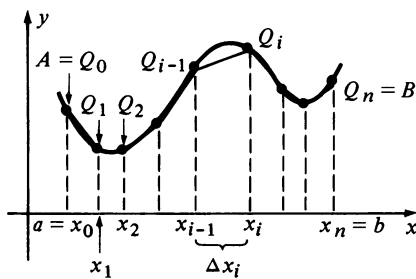


FIGURE 6.38

Applying the Mean Value Theorem (4.12),

$$f(x_i) - f(x_{i-1}) = f'(w_i)(x_i - x_{i-1})$$

where w_i is in the open interval (x_{i-1}, x_i) . Substituting this into the preceding formula and letting $\Delta x_i = x_i - x_{i-1}$, we obtain

$$\begin{aligned} d(Q_{i-1}, Q_i) &= \sqrt{(\Delta x_i)^2 + [f'(w_i) \Delta x_i]^2} \\ &= \sqrt{1 + [f'(w_i)]^2} \Delta x_i. \end{aligned}$$

Consequently,

$$L_P = \sum_{i=1}^n \sqrt{1 + [f'(w_i)]^2} \Delta x_i.$$

Observe that L_P is a Riemann sum for the function g defined by $g(x) = \sqrt{1 + [f'(x)]^2}$. In addition, g is continuous on $[a, b]$ since f' is continuous. As we have mentioned, if the norm $\|P\|$ is small, then the length L_P of the broken line should approximate the length of the graph of f from A to B . Moreover, this approximation should improve as $\|P\|$ decreases. It is natural, therefore, to define the *length* (also called the *arc length*) of the graph of f from A to B as the limit of the sum L_P . Since $g = \sqrt{1 + (f')^2}$ is a continuous function, the limit exists and equals the definite integral $\int_a^b \sqrt{1 + [f'(x)]^2} dx$. This arc length will be denoted by the symbol L_a^b .

Definition (6.15)

Let the function f be smooth on a closed interval $[a, b]$. The **arc length of the graph** of f from $A(a, f(a))$ to $B(b, f(b))$ is given by

$$L_a^b = \int_a^b \sqrt{1 + [f'(x)]^2} dx.$$

Definition (6.15) will be extended to more general graphs in Chapter 13. A graph which has arc length is said to be **rectifiable**. If a function f is defined implicitly by an equation in x and y , then we shall also refer to the *arc length of the graph of the equation*.

Example 1 If $f(x) = 3x^{2/3} - 10$, find the arc length of the graph of f from the point $A(8, 2)$ to $B(27, 17)$.

Solution The graph of f is sketched in Figure 6.39. Since $f'(x) = 2x^{-1/3} = 2/(x^{1/3})$, we have

$$\begin{aligned} L_8^{27} &= \int_8^{27} \sqrt{1 + \left(\frac{2}{x^{1/3}}\right)^2} dx = \int_8^{27} \sqrt{1 + \frac{4}{x^{2/3}}} dx \\ &= \int_8^{27} \frac{\sqrt{x^{2/3} + 4}}{x^{1/3}} dx. \end{aligned}$$

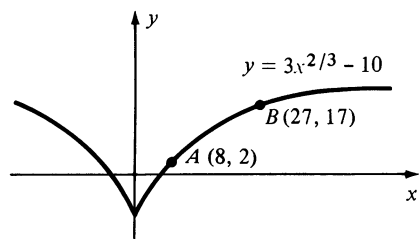


FIGURE 6.39

To evaluate this integral, let

$$u = x^{2/3} + 4 \quad \text{and} \quad du = \frac{2}{3} x^{-1/3} dx = \frac{2}{3} \frac{1}{x^{1/3}} dx.$$

The integral can be expressed in a suitable form for integration by introducing the factor $2/3$ in the integrand and compensating for this by multiplying the integral by $3/2$. Thus

$$L_8^{27} = \frac{3}{2} \int_8^{27} \sqrt{x^{2/3} + 4} \left(\frac{2}{3} \frac{1}{x^{1/3}} \right) dx.$$

If $x = 8$, then $u = (8)^{2/3} + 4 = 8$, whereas if $x = 27$, then $u = (27)^{2/3} + 4 = 13$. Making the substitution and changing the limits of integration,

$$L_8^{27} = \frac{3}{2} \int_8^{13} \sqrt{u} du = u^{3/2} \Big|_8^{13} = 13^{3/2} - 8^{3/2} \approx 24.2. \quad \blacksquare$$

The next result is the analogue of Definition (6.15) for integration with respect to y .

Definition (6.16)

Let the function g be defined by $x = g(y)$, where g is smooth on the interval $[c, d]$. The **arc length of the graph** of g from the point $(g(c), c)$ to the point $(g(d), d)$ is given by

$$L_c^d = \int_c^d \sqrt{1 + [g'(y)]^2} dy.$$

Example 2 Set up an integral for finding the arc length of the graph of $y = y^3 - x$ from $A(0, -1)$ to $B(6, 2)$. Approximate the integral by using Simpson's Rule (5.37) with $n = 6$.

Solution The equation is not of the form $y = f(x)$ and hence Definition (6.15) cannot be applied. However, if we write $x = y^3 - y$, then we can employ Definition (6.16) with $g(y) = y^3 - y$. The graph of the equation is sketched in Figure 6.40.

Using (6.16) with $c = -1$ and $d = 2$ gives us

$$\begin{aligned} L_{-1}^2 &= \int_{-1}^2 \sqrt{1 + (3y^2 - 1)^2} dy \\ &= \int_{-1}^2 \sqrt{9y^4 - 6y^2 + 2} dy. \end{aligned}$$

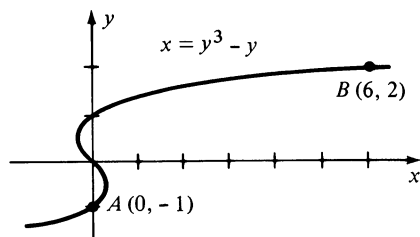


FIGURE 6.40

To use Simpson's Rule, we let $f(y) = \sqrt{9y^4 - 6y^2 + 2}$ and arrange our work as we did in Section 5.7. The reader may verify the entries in the following table.

i	y_i	$f(y_i)$	m	$mf(y_i)$
0	-1.0	2.2361	1	2.2361
1	-0.5	1.0308	4	4.1232
2	0.0	1.4142	2	2.8284
3	0.5	1.0308	4	4.1232
4	1.0	2.2361	2	4.4722
5	1.5	5.8363	4	23.3452
6	2.0	11.0454	1	11.0454

The sum of the numbers in the last column is 52.1737. Hence by Simpson's Rule (5.37) with $a = -1$, $b = 2$, and $n = 6$,

$$\int_{-1}^2 \sqrt{9y^4 - 6y^2 + 2} dy \approx \frac{1}{6} [52.1737] \approx 8.7. \quad \blacksquare$$

If a graph can be decomposed into a finite number of parts, each of which is the graph of a smooth function, then the arc length of the graph is defined as the sum of the arc lengths of the individual graphs. A function of this type is said to be **piecewise smooth** on its domain.

To avoid any misunderstanding in the following discussion, the variable of integration will be denoted by t . In this case the arc length formula given in Definition (6.15) becomes

$$L_a^b = \int_a^b \sqrt{1 + [f'(t)]^2} dt.$$

If f is smooth on $[a, b]$, then f is smooth on $[a, x]$ for every x in $[a, b]$, and the length of the graph from the point $A(a, f(a))$ to the point $Q(x, f(x))$ is given by

$$L_a^x = \int_a^x \sqrt{1 + [f'(t)]^2} dt.$$

If we change the notation and use the symbol $s(x)$ in place of L_a^x , then s may be regarded as a function with domain $[a, b]$, since to each x in $[a, b]$ there corresponds a unique number $s(x)$. We shall call s the *arc length function* for the graph of f , as in the next definition.

Definition (6.17)

Let the function f be smooth on $[a, b]$. The **arc length function** s for the graph of f on $[a, b]$ is given by

$$s(x) = \int_a^x \sqrt{1 + [f'(t)]^2} dt$$

where $a \leq x \leq b$.

As shown in Figure 6.41, the values $s(x)$ of s may be represented geometrically as lengths of arc of the graph of f from $A(a, f(a))$ to $Q(x, f(x))$.

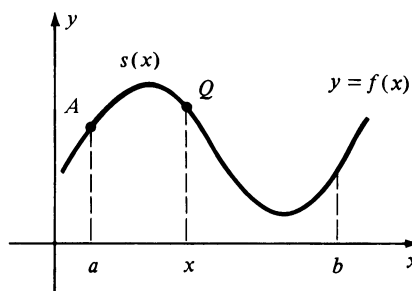


FIGURE 6.41

The following theorem is important for problems involving the arc length function.

Theorem (6.18)

Let f be smooth on $[a, b]$, and let s be the arc length function for the graph of $y = f(x)$ on $[a, b]$. If dx and dy are differentials of x and y , then

$$(i) \quad ds = \sqrt{1 + [f'(x)]^2} dx$$

$$(ii) \quad (ds)^2 = (dx)^2 + (dy)^2.$$

Proof By Definition (6.17) and Theorem (5.25),

$$D_x[s(x)] = D_x \left[\int_a^x \sqrt{1 + [f'(t)]^2} dt \right] = \sqrt{1 + [f'(x)]^2}.$$

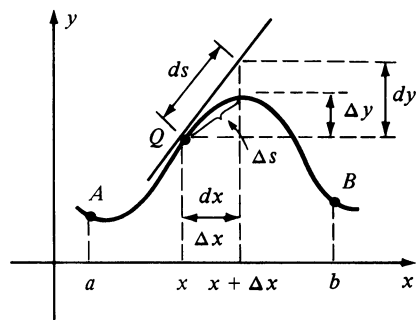
Consequently, by Definition (3.24),

$$ds = s'(x) dx = \sqrt{1 + [f'(x)]^2} dx$$

This proves (i). Squaring both sides of this equation, we obtain

$$\begin{aligned} (ds)^2 &= \{1 + [f'(x)]^2\}(dx)^2 \\ &= (dx)^2 + [f'(x) dx]^2. \end{aligned}$$

Since, by Definition (3.24), $dy = f'(x) dx$, this gives us (ii). $\square$

FIGURE 6.42 $(ds)^2 = (dx)^2 + (dy)^2$

There is an interesting geometric interpretation for formula (ii) of the preceding theorem. Consider $y = f(x)$ and give x an increment Δx . Let Δy denote the change in y and Δs the change in arc length corresponding to Δx . These increments are illustrated in Figure 6.42 where Δy is the amount that the tangent line rises or falls if the independent variable changes from x to $x + \Delta x$ (see also Figure 3.12). Since $(ds)^2 = (dx)^2 + (dy)^2$, ds may be regarded as the length of the hypotenuse of a right triangle with sides $|dx|$

and $|dy|$ as illustrated in Figure 6.42. This figure also indicates that if Δx is small, then ds may be used to approximate the increment Δs of arc length.

Example 3 Use differentials to approximate the arc length of the graph of $y = x^3 + 2x$ from $A(1, 3)$ to $B(1.2, 4.128)$.

Solution If we let $f(x) = x^3 + 2x$, then by (i) of Theorem (6.18),

$$ds = \sqrt{1 + (3x^2 + 2)^2} dx.$$

An approximation may be obtained by letting $x = 1$ and $dx = 0.2$. Thus

$$ds = \sqrt{1 + 5^2}(0.2) = \sqrt{26}(0.2) \approx 1.02. \quad \blacksquare$$

6.7 Exercises

In Exercises 1–4 use Definition (6.15) to find the arc length of the graph of the given equation from A to B .

1 $8x^2 = 27y^3$; $A(1, \frac{2}{3}), B(8, \frac{8}{3})$

2 $(y + 1)^2 = (x - 4)^3$; $A(5, 0), B(8, 7)$

3 $y = 5 - \sqrt{x^3}$; $A(1, 4), B(4, -3)$

4 $y = 6\sqrt[3]{x^2} + 1$; $A(-1, 7), B(-8, 25)$

5–6 Solve Exercises 1 and 2 by means of Definition (6.16).

In Exercises 7–10 find the arc length from A to B of the graph of the given equation.

7 $y = \frac{x^3}{12} + \frac{1}{x}$; $A(1, \frac{13}{12}), B(2, \frac{7}{6})$

8 $y + \frac{1}{4x} + \frac{x^3}{3} = 0$; $A(2, \frac{67}{24}), B(3, \frac{109}{12})$

9 $30xy^3 - y^8 = 15$; $A(\frac{8}{15}, 1), B(\frac{271}{240}, 2)$

10 $x = \frac{y^4}{16} + \frac{1}{2y^2}$; $A(\frac{9}{8}, -2), B(\frac{9}{16}, -1)$

In Exercises 11 and 12 set up (but do not evaluate) the integral for finding the arc length of the graph of the given equation from A to B .

11 $2y^3 - 7y + 2x - 8 = 0$; $A(3, 2), B(4, 0)$

12 $11x - 4x^3 - 7y + 7 = 0$; $A(1, 2), B(0, 1)$

13 Find the arc length of the graph of $x^{2/3} + y^{2/3} = 1$.

14 Find the arc length of the graph of $y = (3x^8 + 5)/30x^3$ from the point $(1, 4/5)$ to the point $(2, 773/240)$.

15 If f is defined by $f(x) = \sqrt[3]{x^2}$, what is the arc length function s ? If x increases from 1 to 1.1, find Δs and ds .

16 Work Exercise 15 if $f(x) = \sqrt{x^3}$.

17 Use differentials to approximate the arc length of the graph of $y = x^2$ from $A(2, 4)$ to $B(2.1, 4.41)$. Illustrate this approximation graphically and compare it with $d(A, B)$.

18 Use differentials to approximate the arc length of the graph of $y + x^3 = 0$ from $A(1, -1)$ to $B(1.1, -1.331)$. Illustrate this approximation graphically and compare it with $d(A, B)$.

19 Use differentials to approximate the arc length of the graph of $y = \cos x$ between the points with x -coordinates $\pi/6$ and $31\pi/180$.

In Exercises 20–22 use Simpson's Rule (5.37) with $n = 4$ to approximate the arc length of the graph of the given equation from A to B .

20 $y = x^2 + x + 3$; $A(-2, 5), B(2, 9)$

21 $y = x^3$; $A(0, 0), B(2, 8)$

22 $y = \sin x$; $A(0, 0), B(\pi/2, 1)$

6.8 Other Applications

In this section we shall consider several miscellaneous applications to give the reader some idea of the versatility of the definite integral. The first three have to do with the study of economics; the remaining applications are from the physical sciences.

It is important for people in business to plan for depreciation of equipment. The most elementary technique employed is the **straight-line method**, in which the rate of depreciation is considered constant. For example, if the depreciation each year is \$200, then the total depreciation $f(t)$ at the end of t years is given by $f(t) = 200t$. Note that the rate of depreciation is $f'(t)$. In many instances the rate of depreciation is not constant. To illustrate, an automobile depreciates very rapidly during the first few years after it is purchased and then more slowly as it gets older. For other items the rate of depreciation may increase from year to year. In general, suppose that the rate of depreciation of a certain piece of equipment over the time interval $[0, t]$ may be approximated by $g(t)$, where g is a continuous function. If the total depreciation in $[0, t]$ is $f(t)$, then $f'(t) = g(t)$ and we may write

$$f(t) = \int_0^t f'(x) dx = \int_0^t g(x) dx.$$

Suppose further that an additional fixed cost C is required to overhaul the equipment after time t . Thus, in the time interval $[0, t]$, the expense $h(t)$ connected with the equipment is given by

$$h(t) = C + \int_0^t g(x) dx.$$

The average expense, that is, the expense per year, is

$$k(t) = \frac{h(t)}{t}.$$

If no other factors are involved, then the best time to overhaul the equipment corresponds to the value of t for which the function k has a relative minimum. This occurs when

$$k'(t) = \frac{th'(t) - h(t)}{t^2} = 0$$

or equivalently, when

$$h'(t) = \frac{h(t)}{t} = k(t).$$

Since $h'(t) = g(t)$, the best time to overhaul occurs when $g(t) = k(t)$, that is, when the rate of depreciation is the same as the average expense.

Example 1 Using the notation of the preceding discussion, suppose that for a given piece of equipment $g(t) = 300\sqrt{t}$ and $C = 500$, where t is in years. When should the equipment be overhauled in order to minimize the average expense?

Solution From the preceding discussion, the average expense is given by

$$\begin{aligned} k(t) &= h(t) \div t = \left[C + \int_0^t g(x) dx \right] \div t \\ &= \left[500 + \int_0^t 300\sqrt{x} dx \right] \div t \\ &= (500 + 200t^{3/2})/t. \end{aligned}$$

From the previous discussion, k will have a minimum value if

$$(500 + 200t^{3/2})/t = 300t^{1/2}.$$

Solving for t we obtain $t = 5^{2/3} \approx 2.924$ or, to the nearest month, 2 years and 11 months. ■

In economics, the process that a corporation uses to increase its accumulated wealth is called **capital formation**. If the amount K of capital at time t can be approximated by $K = f(t)$, where f is a differentiable function, then the rate of change of K with respect to t is called the **net investment flow**. Hence, if I denotes the investment flow, then

$$I = \frac{dK}{dt} = f'(t).$$

Conversely, if I is given by $g(t)$, where the function g is continuous on an interval $[a, b]$, then the increase in capital over this time interval is

$$\int_a^b g(t) dt = f(b) - f(a).$$

Example 2 Suppose a corporation wishes to have its net investment flow approximated by $g(t) = t^{1/3}$, where t is in years and $g(t)$ is in millions of dollars per year. If $t = 0$ corresponds to the present time, estimate the amount of capital formation over the next eight years.

Solution From our discussion, the increase in capital over the next eight years is

$$\int_0^8 g(t) dt = \int_0^8 t^{1/3} dt = \left[\frac{3}{4} t^{4/3} \right]_0^8 = 12.$$

Consequently, the amount of capital formation is \$12,000,000. ■

In many types of employment, a worker must perform the same assignment repeatedly. For example, a boy hired by a bicycle shop may be asked to assemble new bicycles. As he assembles one bicycle after another, the boy should become more proficient and, up to a certain point, should assemble each one in less time than the preceding one. Another example of this process of learning by repetition is that of a key-punch operator who must translate information from written forms to punched cards. The time required to punch each card should decrease as the number of cards increases. As a final illustration, the time required for a person to trace a path through a maze should improve with practice.

Let us consider a general situation in which a certain task is to be repeated many times. Suppose experience has shown that the time required to perform the task for the i th time can be approximated by $f(i)$, where f is a continuous decreasing function on a suitable interval. The total time required to perform the task k times is given by the sum

$$\sum_{i=1}^k f(i) = f(1) + f(2) + \cdots + f(k).$$

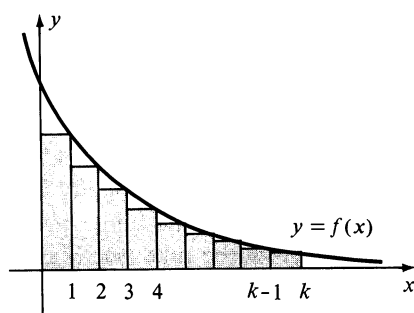


FIGURE 6.43

If we consider the graph of f , then as illustrated in Figure 6.43 the preceding sum equals the area of the pictured inscribed rectangular polygon and, therefore, may be approximated by the definite integral $\int_0^k f(x) dx$. Evidently, the approximation will be close to the actual sum if f decreases slowly on $[0, k]$. If f changes rapidly per unit change in x , then an integral should not be used as an approximation.

Example 3 A company that conducts polls by means of telephone interviews finds that the time required by an employee to complete one interview depends on the number of interviews that the employee has completed previously. Suppose it is estimated that for a certain survey, the number of minutes required to complete the i th interview is given by $f(i) = 6(1 + i)^{-1/5}$. Use a definite integral to approximate the time required for an employee to complete (a) 100 interviews; (b) 200 interviews. If an interviewer receives \$4.80 per hour, estimate how much more expensive it is to have two employees each conduct 100 interviews than it is to have one employee conduct 200 interviews.

Solution As in the preceding discussion, the time required for 100 interviews is approximately

$$\begin{aligned} \int_0^{100} 6(1+x)^{-1/5} dx &= 6 \cdot \frac{5}{4} (1+x)^{4/5} \Big|_0^{100} \\ &= \frac{15}{2} [(101)^{4/5} - 1] \\ &\approx 293.5 \text{ min.} \end{aligned}$$

The time required for 200 interviews is approximately

$$\begin{aligned} \int_0^{200} 6(1+x)^{-1/5} dx &= \frac{15}{2} [(201)^{4/5} - 1] \\ &\approx 514.4 \text{ min.} \end{aligned}$$

Since an interviewer receives \$0.08 per minute, the cost for one employee to conduct 200 interviews is roughly $(\$0.08)(514.4)$ or \$41.15. If two employees each conduct 100 interviews the cost is about $2(\$0.08)(293.5)$ or \$46.96, which is \$5.81 more than the cost of one employee. Note, however, that the time saved in using two people is approximately 220 minutes.

A calculator may be used to show that

$$\sum_{i=1}^{100} 6(1+i)^{-1/5} \approx 291.75$$

and

$$\sum_{i=1}^{200} 6(1+i)^{-1/5} \approx 512.57.$$

Hence the results obtained by integration (the area under the graph of f) are roughly 2 minutes more than the value of the corresponding sum (the area of the inscribed rectangular polygon). ■

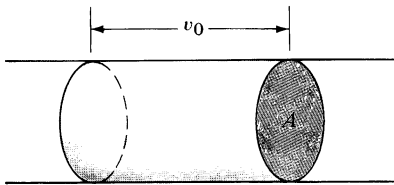


FIGURE 6.44

Our next application of the definite integral is taken from the field of biology. If a liquid flows through a cylindrical tube and if the velocity is a constant v_0 , then the volume of liquid passing a fixed point per unit time is given by $v_0 A$, where A is the area of a cross section of the tube (see Figure 6.44). A more complicated formula is required to study the flow of blood in an arteriole. In this case the flow is in layers, as illustrated in Figure 6.45. In the layer closest to the wall of the arteriole, the blood tends to stick to the wall, and its velocity may be considered zero. The velocity increases as the layers approach the center of the arteriole.

For computational purposes, we may regard the blood flow as consisting of thin cylindrical shells that slide along, with the outer shell fixed and the velocity of the shells increasing as the radii of the shells decrease (see Figure 6.45). If the velocity in each shell is considered constant, then from the theory of liquids in motion, the velocity $v(r)$ in a shell having average radius r is given by

$$v(r) = \frac{P}{4\eta l} (R^2 - r^2)$$

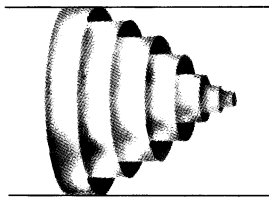


FIGURE 6.45

where R is the radius of the arteriole (in cm), l is the length of the arteriole (in cm), P is the pressure difference between the two ends of the arteriole (in dyn/cm²) and η is the viscosity of the blood (in dyn-sec/cm²). Note that the formula gives zero velocity if $r = R$ and maximum velocity $PR^2/4\eta l$ as r approaches 0. If the radius of the k th shell is r_k and the thickness of the shell is Δr_k , then by (6.6) the volume of blood in this shell is

$$2\pi r_k v(r_k) \Delta r_k = \frac{2\pi r_k P}{4\eta l} (R^2 - r_k^2) \Delta r_k.$$

If there are n shells, then the total flow in the arteriole per unit time may be approximated by

$$\sum_{k=1}^n \frac{2\pi r_k P}{4\eta l} (R^2 - r_k^2) \Delta r_k.$$

To estimate the total flow F , that is, the volume of blood per unit time, we take the limit of this sum as n increases without bound. This leads to the following definite integral:

$$\begin{aligned} F &= \int_0^R \frac{2\pi r P}{4vl} (R^2 - r^2) dr \\ &= \frac{2\pi P}{4vl} \int_0^R (R^2 r - r^3) dr \\ &= \frac{\pi P}{2vl} \left[\frac{1}{2} R^2 r^2 - \frac{1}{4} r^4 \right]_0^R. \end{aligned}$$

Substituting the limits of integration gives us

$$F = \frac{\pi P R^4}{8vl} \text{ cm}^3.$$

This formula for F is not exact since the thickness of the shells cannot be made arbitrarily small. Indeed, the lower limit is the width of a red blood cell, or approximately 2×10^{-4} cm. However, we may assume that the formula gives a reasonable estimate. It is interesting to observe that a small change in the radius of an arteriole produces a large change in the flow, since F is directly proportional to the fourth power of R . A small change in the pressure difference has a lesser effect, since P appears to only the first power.

It should be evident from our work in this chapter that if a quantity can be approximated by a sum of many terms, then it is a candidate for representation as a definite integral. The main requirement is that as the number of terms increases, the sum approaches a limit. Similarly, any quantity that can be interpreted as an area of a region in a plane may be investigated by means of a definite integral. (See, for example, the discussion of hysteresis at the end of Section 6.2.) Conversely, definite integrals allow us to represent physical quantities as areas. Let us conclude this section by reconsidering several earlier concepts from this point of view.

Suppose $v(t)$ is the velocity, at time t , of an object that is moving on a coordinate line. If s is the position function, then $s'(t) = v(t)$, and

$$\int_a^b v(t) dt = \int_a^b s'(t) dt = s(t) \Big|_a^b = s(b) - s(a).$$

If $v(t) > 0$ throughout the time interval $[a, b]$, these equalities tell us that the area under the graph of the function v from a to b is the distance the object travels, as illustrated in Figure 6.46. This observation is useful to an engineer or physicist, who may not have an explicit form for $v(t)$, but merely a graph (or table) indicating the velocity at various times. The distance traveled may then be estimated by approximating the area under the graph.

Incidentally, if $v(t) < 0$ at certain times in $[a, b]$, the graph of v may have the appearance illustrated in Figure 6.47. The figure indicates that the object moved in the negative direction from $t = c$ to $t = d$. The distance it traveled during that time is $\int_c^d |v(t)| dt$. (Why?) It follows that $\int_a^b |v(t)| dt$ is the *total* distance traveled in $[a, b]$, whether $v(t)$ is positive or negative.

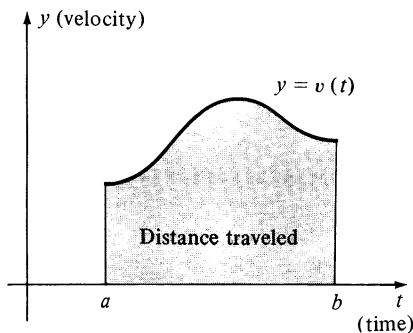


FIGURE 6.46

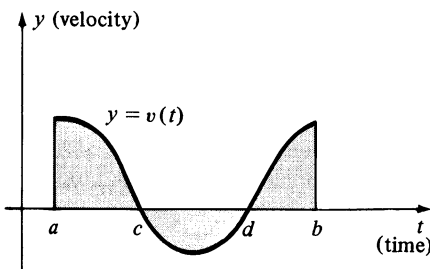


FIGURE 6.47

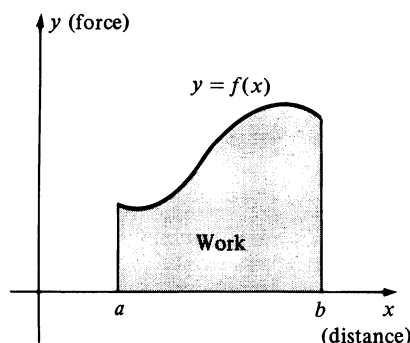


FIGURE 6.48

In Section 6.5 we considered the work W done by a variable force $f(x)$ as its point of application moves along a coordinate line from $x = a$ to $x = b$. The value of W is given by $W = \int_a^b f(x) dx$ (see Definition (6.10)). Suppose $f(x) \geq 0$ for all x on $[a, b]$. If we sketch the graph of f , then, as illustrated in Figure 6.48, the work W is the area under the graph from a to b .

As a final illustration, suppose that the amount of a physical entity such as oil, water, electric power, money supply, bacteria count, blood flow, etc., is increasing or decreasing in some manner, and that $R(t)$ is the rate at which it is changing at time t . If $Q(t)$ is the amount of the entity present at time t , and if Q is differentiable, then we know from Section 4.7 that $Q'(t) = R(t)$. If $R(t) > 0$ (or $R(t) < 0$) in a time interval $[a, b]$, then the amount that the entity increases (or decreases) between $t = a$ and $t = b$ is

$$Q(b) - Q(a) = \int_a^b Q'(t) dt = \int_a^b R(t) dt.$$

As before, this number may be represented as the area of the region in a ty -plane bounded by the graphs of R , $t = a$, $t = b$, and $y = 0$.

Example 4 Starting at 9:00 A.M., oil is pumped into a storage tank at a rate of $150t^{1/2} + 25$ gal/hr, where t is the time (in hours) after 9:00 A.M. How many gallons will have been pumped into the tank at 1:00 P.M.?

Solution Letting $R(t) = 150t^{1/2} + 25$ in the preceding discussion, we obtain

$$\begin{aligned} \int_0^4 (150t^{1/2} + 25) dt &= 100t^{3/2} + 25t \Big|_0^4 \\ &= 100(4)^{3/2} + 25(4) \\ &= 100(8) + 100 = 900 \text{ gal.} \end{aligned}$$

We have given only a few illustrations of the use of definite integrals. The interested reader may find many more in books on physics and engineering.

6.8 Exercises

- 1 The rate of depreciation of a certain piece of equipment over the time interval $[0, 3]$ may be approximated by $g(t) = 1 - (t^2/9)$ where t is in years and $g(t)$ is in hundreds of dollars. Find the total depreciation at the end of (a) 6 months; (b) 1 year; (c) 18 months; (d) 2 years.
- 2 The rate of depreciation of a certain piece of equipment over the time interval $[0, 6]$ may be approximated by

$g(t) = \sqrt{10 - t}$, where t is in years and $g(t)$ is in thousands of dollars. If, after six years, an additional cost of \$420 is needed to overhaul the equipment, what is the average expense for the six years?

- 3 If, in Example 1, $g(t) = 100t^{2/3}$ and $C = 400$, find the best time to overhaul the equipment.

- 4 If, in Example 2, the rate of investment is approximated by $g(t) = 2t(3t + 1)$, where $g(t)$ is in thousands of dollars, find the amount of capital formation over the intervals $[0, 5]$ and $[5, 10]$.
- 5 A key-punch operator is required to transfer registration data of college students from written sheets to punched cards. It is estimated that the number of minutes required to punch the i th card is approximately $f(i) = 6(1 + i)^{-1/3}$. Use a definite integral to estimate the time required for (a) one operator to punch 600 cards; (b) two operators to punch 300 cards each.
- 6 It is estimated that the number of minutes needed for a person to trace a path through a certain maze without error is given by $f(i) = 5i^{-1/2}$, where i is the number of trials previously completed. Approximately how much time is required to complete 10 trials?
- 7 A small parts manufacturer estimates that the time required for a worker to assemble a certain item depends on the number of items the worker has previously assembled. If the time (in minutes) required to assemble the n th item is given by $f(n) = 20(n + 1)^{-0.4} + 3$, approximate the time required to assemble (a) 1 item; (b) 4 items; (c) 8 items; (d) 16 items.
- 8 Use a definite integral to approximate the sum
- $$\sum_{k=1}^{100} k(k^2 + 1)^{-1/4}.$$
- 9 A motorboat uses gasoline at the rate of $t\sqrt{9 - t^2}$ gal/hr. If the motor is started at $t = 0$, how much gasoline is used in the next two hours?
- 10 The population of a city has increased since 1980 at a rate of $1.5 + 0.3\sqrt{t} + 0.006t^2$ thousand people per year, where t is the number of years after 1980. Assuming that this rate continues and that the population was 50,000 in 1980, estimate the population in 1989.

- 11 The accompanying figure indicates the velocity of an automobile (in mi/hr) as it traveled along a freeway over a twelve minute interval. Approximate the distance traveled.

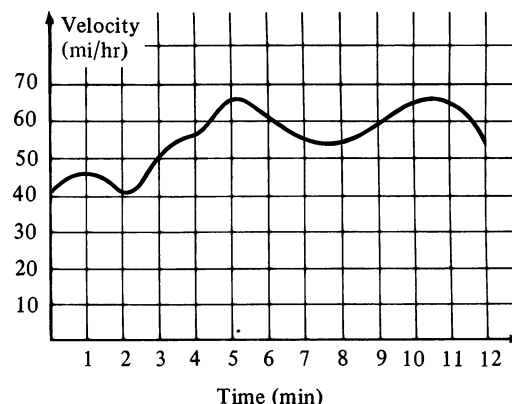


FIGURE FOR EXERCISE 11

- 12 The acceleration (in ft/sec²) of an automobile over a period of 8 seconds is indicated in the accompanying figure. Approximate the net change in velocity in this time period.

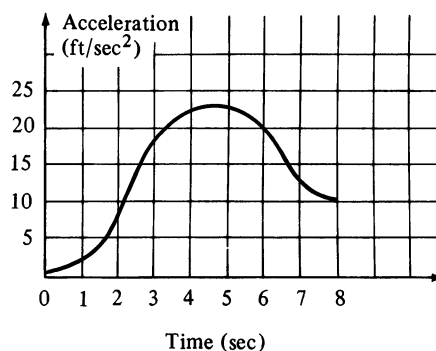


FIGURE FOR EXERCISE 12

6.9 Review: Concepts

Define or discuss each of the following.

- 1 The area between the graphs of two continuous functions
- 2 Solid of revolution
- 3 Methods of finding volumes of solids of revolution
- 4 Volumes by slicing
- 5 Work

- 6 Force exerted by a liquid
- 7 Smooth function
- 8 Arc length of a graph
- 9 Piecewise smooth function
- 10 The arc length function

Review: Exercises

In Exercises 1 and 2, sketch the region bounded by the graphs of the given equations and find the area in two ways, first by integrating with respect to x , and second by integrating with respect to y .

1 $y = -x^2$, $y = x^2 - 8$

2 $y^2 = 4 - x$, $x + 2y - 1 = 0$

In Exercises 3 and 4 find the area of the region bounded by the graphs of the given equations.

3 $x = y^2$, $x + y = 1$

4 $y + x^3 = 0$, $y = \sqrt{x}$, $3y + 7x - 10 = 0$

5 Find the area of the region between the graphs of the equations $y = \cos(x/2)$ and $y = \sin x$ from $x = \pi/3$ to $x = \pi$.

6 The region bounded by the graph of $y = \sqrt{1 + \cos 2x}$ and the x -axis, from $x = 0$ to $x = \pi/2$, is revolved about the x -axis. Find the volume of the resulting solid.

In Exercises 7–10 sketch the region R bounded by the graphs of the given equations and find the volume of the solid generated by revolving R about the indicated axis.

7 $y = \sqrt{4x + 1}$, $y = 0$, $x = 0$, $x = 2$; about the x -axis

8 $y = x^4$, $y = 0$, $x = 1$; about the y -axis

9 $y = x^3 + 1$, $x = 0$, $y = 2$; about the y -axis

10 $y = \sqrt[3]{x}$, $y = \sqrt{x}$; about the x -axis

11 Find the volume of the solid generated by revolving the region bounded by the graphs of the equations $y = 4x^2$ and $4x + y - 8 = 0$ (a) about the x -axis; (b) about the line $x = 1$; (c) about the line $y = 16$.

12 Find the volume of the solid generated by revolving the region bounded by the graphs of $y = x^3$, $x = 2$, and $y = 0$ (a) about the x -axis; (b) about the y -axis; (c) about the

line $x = 2$; (d) about the line $x = 3$; (e) about the line $y = 8$; (f) about the line $y = -1$.

13 Find the arc length of the graph of $(x + 3)^2 = 8(y - 1)^3$ from the point $A(-2, \frac{3}{2})$ to the point $B(5, 3)$.

14 A solid has, for its base, the region in the xy -plane bounded by the graphs of $y^2 = 4x$ and $x = 4$. Find the volume of the solid if every cross section by a plane perpendicular to the x -axis is an isosceles right triangle with one of the equal sides on the base of the solid.

15 An above-ground swimming pool has the shape of a right circular cylinder of diameter 12 ft and height 5 ft. If the depth of the water in the pool is 4 ft, find the work required to empty the pool by pumping the water out over the top.

16 As a bucket is raised a distance of 30 ft from the bottom of a well, water leaks out at a uniform rate. Find the work done if the bucket originally contains 24 lb of water and one-third leaks out. Assume that the weight of the empty bucket is 4 lb and neglect the weight of the rope.

17 A square plate of side 4 ft is submerged vertically in water such that one of the diagonals is parallel to the surface. If the distance from the surface to the center of the plate is 6 ft, find the force exerted by the water on one side of the plate.

18 Use differentials to approximate the arc length of the graph of $y = 2 \sin(x/3)$ between the points with x -coordinates π and $91\pi/90$.

If $\lim_{\|P\| \rightarrow 0} \sum_i \pi w_i^4 \Delta x_i$ represents the limit of a sum for a function f on the interval $[0, 1]$, solve Exercises 19–22.

19 Find the value of the limit.

20 Interpret the limit as the area of a region in the xy -plane.

21 Interpret the limit as the volume of a solid of revolution.

22 Interpret the limit as the work done by a force.

7 Exponential and Logarithmic Functions

Most of the functions considered in preceding chapters are algebraic functions. Functions that are not algebraic are called *transcendental*. The sine and cosine functions are examples of transcendental functions. In this chapter we introduce two other important transcendental

functions and investigate some of their properties. Since the two functions to be defined are inverse functions of one another, we shall begin the chapter by discussing this fundamental concept.

7.1 Inverse Functions

In Section 1.5 we defined the composite function $f \circ g$ of two functions f and g by means of the formula $(f \circ g)(x) = f(g(x))$. Similarly, $(g \circ f)(x) = g(f(x))$. Usually $f(g(x))$ and $g(f(x))$ are not the same, that is, $f \circ g \neq g \circ f$. In certain cases it may happen that equality *does* occur. Of major importance is the case in which $f(g(x))$ and $g(f(x))$ are not only identical, but both are equal to x . Needless to say, f and g must be very special functions for this to happen. In the following discussion we indicate the manner in which they will be restricted.

Suppose f is a *one-to-one function* with domain X and range Y (see Definition (1.22)). This implies that each element y of Y is the image of precisely one element x of X . We can also say that *each element y of Y can be written in one and only one way in the form $f(x)$, where x is in X* . We may then define a function g from Y to X by demanding that

$$g(y) = g(f(x)) = x \quad \text{for every } x \text{ in } X.$$

This amounts to *reversing* the correspondence given by f . If f is represented geometrically by drawing arrows as in (i) of Figure 7.1, then g can be represented by simply *reversing* these arrows as illustrated in (ii) of the figure.

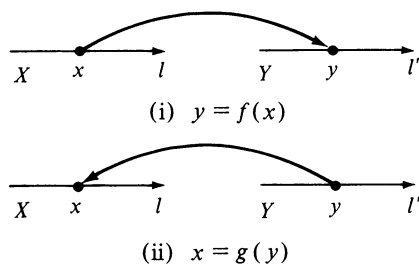


FIGURE 7.1

As illustrated in Figure 7.1, if $f(x) = y$, then $x = g(y)$. This implies that

$$f(g(y)) = y \quad \text{for every } y \text{ in } Y.$$

Since the notation used for the variable is immaterial, we may write

$$f(g(x)) = x \quad \text{for every } x \text{ in } Y.$$

The functions f and g are called *inverse functions* of one another, according to the following definition.

Definition (7.1)

If f is a one-to-one function with domain X and range Y , then a function g with domain Y and range X is called the **inverse function of f** if

$$f(g(x)) = x \quad \text{for every } x \text{ in } Y$$

and

$$g(f(x)) = x \quad \text{for every } x \text{ in } X.$$

If f is one-to-one there is only one inverse function g of f and, moreover, g is one-to-one (see Exercises 16 and 18). It follows from Definition (7.1) that f is the inverse function of g . We often say that f and g are *inverse functions of one another*. The following relationship (illustrated in Figure 7.1) always holds between a one-to-one function f and its inverse function g :

$$(7.2) \quad y = f(x) \quad \text{if and only if} \quad x = g(y).$$

If g is the inverse function of f , then by Definition (7.1), $(g \circ f)(x) = x$ for every x in X , that is, $g \circ f$ is the identity function on X . Similarly, since $(f \circ g)(x) = x$, for all x in Y , $f \circ g$ is the identity function on Y . For this reason the symbol f^{-1} is often used to denote the inverse function of f . Employing this notation,

$$f^{-1}(f(x)) = x \quad \text{for every } x \text{ in } X$$

$$f(f^{-1}(x)) = x \quad \text{for every } x \text{ in } Y.$$

The symbol -1 used here should not be mistaken for an exponent; that is, $f^{-1}(x)$ does not mean $1/f(x)$. The reciprocal $1/f(x)$ may be denoted by $[f(x)]^{-1}$.

Inverse functions are very important in the study of trigonometry. In this chapter we shall discuss several other important inverse functions. It is important to remember that in order to define the inverse of a function f , it is absolutely essential that f be one-to-one. The most common examples of one-to-one functions are those that are increasing or decreasing on their domains, for in this case, if $a \neq b$ in the domain, then $f(a) \neq f(b)$ in the range.

An algebraic method can sometimes be used to find the inverse of a one-to-one function f with domain X and range Y . Given any x in X , its image y in Y may be found by means of the equation $y = f(x)$. To determine the inverse function g , we wish to *reverse* this procedure, in the sense that given y , the

element x may be found. Since x and y are related by means of $y = f(x)$, it follows that if this equation can be solved for x in terms of y , we may arrive at the inverse function g . This technique is illustrated in the following examples.

Example 1 If $f(x) = 3x - 5$ for every real number x , find the inverse function of f .

Solution The graph of the given linear function f is a line of slope 3, and hence f is increasing for all x . It follows that f is a one-to-one function with domain and range $\mathbb{R}$, and hence the inverse function g exists. If we let

$$y = 3x - 5$$

and then solve for x in terms of y , we obtain

$$x = \frac{y + 5}{3}.$$

The last equation enables us to find x when given y . Letting

$$g(y) = \frac{y + 5}{3}$$

gives us a function g from Y to X that reverses the correspondence determined by f . Since the symbol used for the independent variable is immaterial, we may replace y by x in the expression for g , obtaining

$$g(x) = \frac{x + 5}{3}.$$

To verify that g is actually the inverse function of f , we must verify that the two conditions of Definition (7.1), $f(g(x)) = x$ and $g(f(x)) = x$, are fulfilled. Thus,

$$f(g(x)) = f\left(\frac{x + 5}{3}\right) = 3\left(\frac{x + 5}{3}\right) - 5 = x.$$

Similarly,

$$g(f(x)) = g(3x - 5) = \frac{(3x - 5) + 5}{3} = x.$$

This proves that g is the inverse function of f . Using the f^{-1} notation,

$$f^{-1}(x) = \frac{x + 5}{3}. \quad \blacksquare$$

Example 2 Find the inverse function of f if the domain X is the interval $[0, \infty)$ and $f(x) = x^2 - 3$ for all x in X .

Solution The domain has been restricted so that f is increasing and hence is one-to-one. The range of f is the interval $[-3, \infty)$. As in Example 1 we begin by considering the equation

$$y = x^2 - 3.$$

Solving for x gives us

$$x = \pm\sqrt{y+3}.$$

Since x is nonnegative, we reject $x = -\sqrt{y+3}$ and, as in the preceding example, we let

$$g(y) = \sqrt{y+3} \quad \text{or equivalently,} \quad g(x) = \sqrt{x+3}.$$

We now check the two conditions in Definition (7.1), obtaining

$$f(g(x)) = f(\sqrt{x+3}) = (\sqrt{x+3})^2 - 3 = (x+3) - 3 = x$$

$$\text{and} \quad g(f(x)) = g(x^2 - 3) = \sqrt{(x^2 - 3) + 3} = x.$$

This proves that

$$f^{-1}(x) = \sqrt{x+3}, \quad \text{where } x \geq -3. \quad \blacksquare$$

There is an interesting relationship between the graphs of a function f and its inverse function f^{-1} . We first note that f maps a into b if and only if f^{-1} maps b into a ; that is, $b = f(a)$ means the same thing as $a = f^{-1}(b)$. These equations imply that the point (a, b) is on the graph of f if and only if the point (b, a) is on the graph of f^{-1} . As an illustration, in Example 2 we found that the functions f and f^{-1} given by

$$f(x) = x^2 - 3 \quad \text{and} \quad f^{-1}(x) = \sqrt{x+3}$$

are inverse functions of one another, provided x is suitably restricted. Some points on the graph of f are $(0, -3)$, $(1, -2)$, $(2, 1)$, and $(3, 6)$. Corresponding points on the graph of f^{-1} are $(-3, 0)$, $(-2, 1)$, $(1, 2)$, and $(6, 3)$. The graphs of f and f^{-1} are sketched on the same coordinate axes in Figure 7.2. If the page is folded along the line $y = x$, which bisects quadrants I and III (as indicated by the dashes in the figure), then the graphs of f and f^{-1} coincide. The two graphs are said to be *reflections* of one another through the line l with equation $y = x$ (or *symmetric* with respect to l). This is typical of the graphs of all functions f that have inverse functions f^{-1} .

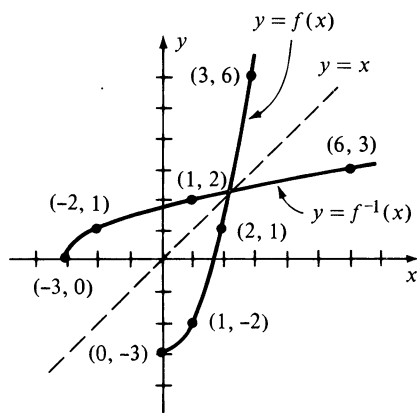


FIGURE 7.2

7.1 Exercises

In Exercises 1–4 prove that f and g are inverse functions of one another.

1 $f(x) = 9x + 2$, $g(x) = \frac{1}{9}x - \frac{2}{9}$

2 $f(x) = x^3 + 1$, $g(x) = \sqrt[3]{x-1}$

3 $f(x) = \sqrt{2x+1}$, $x \geq -\frac{1}{2}$; $g(x) = \frac{1}{2}x^2 - \frac{1}{2}$, $x \geq 0$

4 $f(x) = 1/(x-1)$, $x > 1$; $g(x) = (1+x)/x$, $x > 0$

In Exercises 5–14 find the inverse function of f .

5 $f(x) = 8 + 11x$

6 $f(x) = 1/(8 + 11x)$, $x > -\frac{8}{11}$

7 $f(x) = 6 - x^2$, $0 \leq x \leq \sqrt{6}$

8 $f(x) = 2x^3 - 5$

9 $f(x) = \sqrt{7x-2}$, $x \geq \frac{2}{7}$

- 10 $f(x) = \sqrt{1 - 4x^2}, 0 \leq x \leq \frac{1}{2}$
- 11 $f(x) = 7 - 3x^3$
- 12 $f(x) = x$
- 13 $f(x) = (x^3 + 8)^5$
- 14 $f(x) = x^{1/3} + 2$
- 15 Sketch the graphs of f and g given in Exercise 1 on the same coordinate plane. Do the same for the functions defined in Exercise 3.
- 16 If f is a one-to-one function with domain X and range Y , and if g is the inverse function of f , prove that g is a one-to-one function with domain Y and range X .
- 17 Prove that the linear function f defined by $f(x) = ax + b$ has an inverse function if $a \neq 0$. Does a constant function have an inverse? Does an identity function have an inverse?
- 18 Prove that a function f can have at most one inverse function.
- 19 Prove that not every polynomial function has an inverse function.
- 20 If f has an inverse function f^{-1} , prove that f^{-1} has an inverse function and that $(f^{-1})^{-1} = f$.

7.2 The Natural Logarithmic Function

If a is a positive real number, it is easy to define a^n for every integral or rational exponent n . However, the extension to irrational exponents requires more complex concepts than those discussed in elementary mathematics courses. To give meaning to a symbol such as a^π , we could express π as a nonterminating decimal $3.14159\dots$ and consider the numbers $a^3, a^{3.1}, a^{3.14}, a^{3.141}, a^{3.1415}, a^{3.14159}$, etc. Since we expect successive powers to get closer to a^π , it would seem natural to define a^π as $\lim_{r \rightarrow \pi} a^r$, where r is restricted to *rational* numbers. Unfortunately, there are several disadvantages to this technique. In addition to showing that the limit exists and is unique, we would also have to prove that laws of exponents such as $a^u a^v = a^{u+v}$ are true for all real numbers u and v , a formidable task, to say the least.

Irrational exponents are sometimes defined by employing the Completeness Property for real numbers discussed in Chapter 11 (see (11.9)). This method is rigorous, but it requires a great deal of painstaking work to establish properties of exponents.

Due to the difficulties we have mentioned, it is customary, in precalculus courses, to *assume* that if $a > 0$, then a^u exists for every real number u , and that the usual laws of exponents are true for real exponents. Logarithms with base a are introduced by demanding that

$$u = \log_a v \quad \text{if and only if} \quad a^u = v.$$

Properties of logarithms may then be obtained using the (assumed) laws of exponents. Although this development is suitable in elementary algebra, it is inappropriate in advanced courses, where the standards of mathematical rigor are higher.

In this chapter we shall use concepts developed in calculus to state definitions of $\log_a x$ for every positive real number x and of a^x for every real number x . Our approach to these definitions will be first to use a definite integral to introduce the *natural logarithmic function*, denoted by $\ln$. Later, the natural logarithmic function will be used to define the *natural exponential*

function. Finally, we will give meaning to the expressions a^x and $\log_a x$. The reason for proceeding in this fashion is that it provides a precise method of defining general logarithmic and exponential functions without invoking specialized limiting processes. Our technique will also allow us to establish results on continuity, derivatives, and integrals in a very simple manner. As will be seen, the end result will be the laws involving exponents and logarithms that students encounter before taking calculus.

If a function f is continuous on a closed interval $[a, b]$, then as in the proof of Theorem (5.21) we can define a function F by

$$F(x) = \int_a^x f(t) dt$$

where x is any number in $[a, b]$. If $f(t) \geq 0$ throughout $[a, b]$, then $F(x)$ equals the area under the graph of f from a to x . Let us consider the special case $f(t) = t^n$, where n is any integer different from -1 . By the Fundamental Theorem of Calculus,

$$\begin{aligned} F(x) &= \int_a^x t^n dt = \frac{1}{n+1} t^{n+1} \Big|_a^x \\ &= \frac{1}{n+1} (x^{n+1} - a^{n+1}) \end{aligned}$$

provided t^n is defined throughout $[a, x]$. It is necessary to exclude $f(t) = 1/t$ in the last formula since $1/(n+1)$ is undefined if $n = -1$. Indeed, up to this stage of our work it has been impossible to find an antiderivative for $f(x) = 1/x$. We shall now remedy this situation by introducing a function whose derivative is $1/x$.

Definition (7.3)

The **natural logarithmic function**, denoted by **ln**, is defined by

$$\ln x = \int_1^x \frac{1}{t} dt$$

for all $x > 0$.

The expression $\ln x$ is called the **natural logarithm of x** . The restriction $x > 0$ is necessary since $\int_1^x (1/t) dt$ does not exist if $x \leq 0$. (Why?) The reason that the term *logarithmic* is used in Definition (7.3) will become clear after we have proved that $\ln$ satisfies the laws of logarithms studied in precalculus courses (see Theorem (7.6)).

It follows from Theorem (5.25) that

$$D_x \int_1^x \frac{1}{t} dt = \frac{1}{x}$$

for all $x > 0$ (see Example 6 of Section 5.5). This gives us the next theorem.

Theorem (7.4)

$$D_x(\ln x) = \frac{1}{x}$$

Thus, $\ln x$ is an antiderivative of $1/x$. Since $\ln x$ is differentiable, and its derivative $1/x$ is positive for all $x > 0$, it follows from Theorems (3.8) and (4.13) that *the natural logarithmic function is continuous and increasing throughout its domain.*

Let us sketch the graph of $y = \ln x$. The y -coordinates of points on the graph are given by the integral in Definition (7.3). In particular, if $x = 1$, then by Definition (5.10),

$$\ln 1 = \int_1^1 \frac{1}{t} dt = 0.$$

Thus the graph of $y = \ln x$ has x -intercept 1. Since $\ln$ is an increasing function, it follows that if $x > 1$, then the point (x, y) on the graph lies above the x -axis, whereas if $0 < x < 1$, then (x, y) lies below the x -axis. To estimate y if $x \neq 1$ we may apply either the Trapezoidal Rule or Simpson's Rule. If $x = 2$, then by Example 2 in Section 5.7,

$$\ln 2 = \int_1^2 \frac{1}{t} dt \approx 0.6932 \approx 0.69.$$

It will be shown in Theorem (7.6) that if $a > 0$, then $\ln a^r = r \ln a$ for every rational number r . Using this result, we obtain

$$\ln 4 = \ln 2^2 = 2 \ln 2 \approx 2(0.69) = 1.38$$

$$\ln 8 = \ln 2^3 = 3 \ln 2 \approx 2.07$$

$$\ln \frac{1}{2} = \ln 2^{-1} = -\ln 2 \approx -0.69$$

$$\ln \frac{1}{4} = \ln 2^{-2} = -2 \ln 2 \approx -1.38$$

$$\ln \frac{1}{8} = \ln 2^{-3} = -3 \ln 2 \approx -2.07.$$

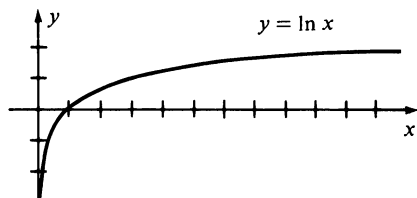


FIGURE 7.3

Table C provides a list of natural logarithms of many other numbers correct to three decimal places. Calculators may also be used to estimate values of $\ln$.

Plotting the points that correspond to the coordinates we have found and using the fact that $\ln$ is continuous and increasing gives us the sketch in Figure 7.3.

At the end of this section it will be proved that

$$\lim_{x \rightarrow \infty} \ln x = \infty$$

that is, the values of $\ln$ can be made arbitrarily large by choosing x sufficiently large. It will also be shown that

$$\lim_{x \rightarrow 0^+} \ln x = -\infty$$

which means that the y -axis is a vertical asymptote for the graph.

Finally, to investigate concavity we note that

$$D_x^2(\ln x) = D_x\left(\frac{1}{x}\right) = -\frac{1}{x^2}$$

which is negative for every $x > 0$. Hence by (4.16), the graph of the natural logarithmic function is concave downward at every point $P(c, \ln c)$ on the graph.

The next result generalizes Theorem (7.4).

Theorem (7.5)

If $y = \ln u$ and $u = g(x)$, where g is differentiable and $g(x) > 0$, then

$$D_x \ln u = \frac{1}{u} D_x u.$$

Proof

By the Chain Rule,

$$\frac{dy}{dx} = \frac{dy}{du} \frac{du}{dx} = \frac{1}{u} D_x u.$$

□

Note that if $u = x$, then Theorem (7.5) reduces to (7.4).

Example 1

(a) Find $f'(x)$ if $f(x) = \ln(x^2 + 6)$.

(b) Find y' if $y = \ln \sqrt{x+1}$, where $x > -1$.

Solution

(a) Letting $u = x^2 + 6$ in Theorem (7.5), we obtain

$$f'(x) = D_x \ln(x^2 + 6) = \frac{1}{x^2 + 6} D_x(x^2 + 6) = \frac{2x}{x^2 + 6}.$$

(b) Letting $u = \sqrt{x+1}$ in Theorem (7.5) gives us

$$\begin{aligned} y' &= \frac{1}{\sqrt{x+1}} D_x \sqrt{x+1} = \frac{1}{\sqrt{x+1}} \cdot \frac{1}{2} (x+1)^{-1/2} \\ &= \frac{1}{\sqrt{x+1}} \cdot \frac{1}{2} \frac{1}{\sqrt{x+1}} = \frac{1}{2(x+1)}. \end{aligned}$$

■

The next theorem shows that natural logarithms obey the same laws of logarithms studied in precalculus mathematics courses.

Theorem (7.6)

If $p > 0$ and $q > 0$, then

$$(i) \quad \ln pq = \ln p + \ln q.$$

$$(ii) \quad \ln \left(\frac{p}{q} \right) = \ln p - \ln q.$$

$$(iii) \quad \ln p^r = r \ln p, \text{ where } r \text{ is any rational number.}$$

Proof

(i) If $p > 0$, then using Theorem (7.5) with $u = px$ gives us

$$D_x \ln (px) = \frac{1}{px} D_x (px) = \frac{1}{px} p = \frac{1}{x}.$$

Since $\ln px$ has the same derivative as $\ln x$ for all $x > 0$, it follows from Theorem (4.32) that these expressions differ by a constant; that is,

$$\ln px = \ln x + C$$

for some real number C . Substituting 1 for x we obtain

$$\ln p = \ln 1 + C.$$

Since $\ln 1 = 0$ we see that $C = \ln p$, and hence

$$\ln px = \ln x + \ln p.$$

Substituting q for x in the last equation gives us

$$\ln pq = \ln q + \ln p$$

which is what we wished to prove.

(ii) Using the formula established in part (i), with $p = 1/q$, we see that

$$\ln \frac{1}{q} + \ln q = \ln \left(\frac{1}{q} \cdot q \right) = \ln 1 = 0$$

and hence

$$\ln \frac{1}{q} = -\ln q.$$

Consequently,

$$\ln \frac{p}{q} = \ln \left(p \cdot \frac{1}{q} \right) = \ln p + \ln \frac{1}{q} = \ln p - \ln q.$$

(iii) If r is any rational number and $x > 0$, then by Theorem (7.5) with $u = x^r$,

$$D_x (\ln x^r) = \frac{1}{x^r} D_x (x^r) = \frac{1}{x^r} r x^{r-1} = r \left(\frac{1}{x} \right).$$

However, by Theorems (3.14) and (7.4), we may also write

$$D_x(r \ln x) = r D_x(\ln x) = r \left(\frac{1}{x} \right).$$

Consequently, $D_x(\ln x^r) = D_x(r \ln x)$.

Since $\ln x^r$ and $r \ln x$ have the same derivative, it follows from Theorem (4.32) that

$$\ln x^r = r \ln x + C$$

for some constant C . If we let $x = 1$ in the last formula we obtain

$$\ln 1 = r \ln 1 + C.$$

Since $\ln 1 = 0$, this implies that $C = 0$ and, therefore,

$$\ln x^r = r \ln x.$$

In Section 7.5 we shall extend this law to irrational exponents. □

The next example illustrates the fact that it is sometimes convenient to use Theorem (7.6) before differentiating.

Example 2 Find $f'(x)$ if $f(x) = \ln [\sqrt{6x-1}(4x+5)^3]$, where $x > \frac{1}{6}$.

Solution We first write $\sqrt{6x-1} = (6x-1)^{1/2}$ and then use (i) and (iii) of Theorem (7.6), obtaining

$$\begin{aligned} f(x) &= \ln [(6x-1)^{1/2}(4x+5)^3] \\ &= \ln (6x-1)^{1/2} + \ln (4x+5)^3 \\ &= \frac{1}{2} \ln (6x-1) + 3 \ln (4x+5). \end{aligned}$$

We may now differentiate with ease, using Theorem (7.5). Thus,

$$\begin{aligned} f'(x) &= \frac{1}{2} \cdot \frac{1}{6x-1} (6) + 3 \cdot \frac{1}{4x+5} (4) \\ &= \frac{3}{6x-1} + \frac{12}{4x+5} \\ &= \frac{84x+3}{(6x-1)(4x+5)}. \end{aligned} \quad \blacksquare$$

In future examples and exercises, if a function is defined in terms of the natural logarithmic function, its domain will not usually be stated explicitly. Instead it will be assumed that x is restricted to values for which the given expression has meaning.

Example 3 Find y' if $y = \ln \sqrt[3]{\frac{x^2 - 1}{x^2 + 1}}$.

Solution We first use Theorem (7.6) to change the form of y as follows:

$$\begin{aligned} y &= \ln \left(\frac{x^2 - 1}{x^2 + 1} \right)^{1/3} = \frac{1}{3} \ln \left(\frac{x^2 - 1}{x^2 + 1} \right) \\ &= \frac{1}{3} [\ln(x^2 - 1) - \ln(x^2 + 1)]. \end{aligned}$$

Next we apply Theorem (7.5), obtaining

$$\begin{aligned} y' &= \frac{1}{3} \left[\frac{1}{x^2 - 1} (2x) - \frac{1}{x^2 + 1} (2x) \right] \\ &= \frac{2x}{3} \left[\frac{1}{x^2 - 1} - \frac{1}{x^2 + 1} \right] \\ &= \frac{2x}{3} \left[\frac{2}{(x^2 - 1)(x^2 + 1)} \right] = \frac{4x}{3(x^4 - 1)}. \end{aligned}$$

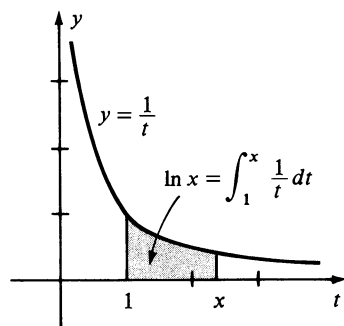


FIGURE 7.4

We shall conclude this section by investigating the variation of $\ln x$ as $x \rightarrow \infty$ and as $x \rightarrow 0^+$.

If $x > 1$ and t is in the interval $[1, x]$, then $1/t > 0$, and hence we may interpret the integral $\int_1^x (1/t) dt = \ln x$ as the area of the region in a ty -coordinate system bounded by the graphs of $y = 1/t$, $t = 1$, $t = x$, and the t -axis. A region of this type is illustrated in Figure 7.4.

Next, observe that the sum of the areas of the three rectangles shown in Figure 7.5 is

$$\frac{1}{2} + \frac{1}{3} + \frac{1}{4} = \frac{13}{12}.$$

Since the area under the graph of $y = 1/t$ from $t = 1$ to $t = 4$ is $\ln 4$ we see that

$$\ln 4 > \frac{13}{12} > 1.$$

It follows that if M is any positive rational number, then

$$M \ln 4 > M, \quad \text{or} \quad \ln 4^M > M.$$

If $x > 4^M$, then since $\ln$ is an increasing function,

$$\ln x > \ln 4^M > M.$$

This proves that $\ln x$ can be made as large as desired by choosing x sufficiently large, that is,

$$\lim_{x \rightarrow \infty} \ln x = \infty.$$

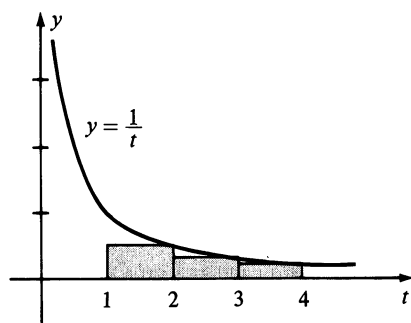


FIGURE 7.5

Next we note that

$$-\ln \frac{1}{x} = -(\ln 1 - \ln x) = \ln x$$

and hence

$$\lim_{x \rightarrow 0^+} \ln x = \lim_{x \rightarrow 0^+} \left(-\ln \frac{1}{x} \right).$$

As x approaches zero through positive values, $1/x$ becomes positively infinite and, therefore, so does $\ln(1/x)$. Consequently, $-\ln(1/x)$ becomes negatively infinite, that is,

$$\lim_{x \rightarrow 0^+} \ln x = -\infty.$$

7.2 Exercises

In Exercises 1–28 find $f'(x)$ if $f(x)$ equals the given expression.

- | | |
|---|--|
| 1 $\ln(9x + 4)$ | 2 $\ln(x^4 + 1)$ |
| 3 $\ln(2 - 3x)^5$ | 4 $\ln(5x^2 + 1)^3$ |
| 5 $\ln \sqrt{7 - 2x^3}$ | 6 $\ln \sqrt[3]{6x + 7}$ |
| 7 $\ln(3x^2 - 2x + 1)$ | 8 $\ln(4x^3 - x^2 + 2)$ |
| 9 $\ln \sqrt[3]{4x^2 + 7x}$ | 10 $\ln x $ |
| 11 $x \ln x$ | 12 $x^2/(\ln x)$ |
| 13 $\ln \sqrt{x} + \sqrt{\ln x}$ | 14 $\ln x^3 + (\ln x)^3$ |
| 15 $\frac{1}{\ln x} + \ln \left(\frac{1}{x} \right)$ | 16 $\ln \frac{\sqrt{4 + x^2}}{4 - x^2}$ |
| 17 $\ln(5x - 7)^4(2x + 3)^3$ | 18 $\ln \sqrt[3]{4x - 5}(3x + 8)^2$ |
| 19 $\ln \frac{\sqrt{x^2 + 1}}{(9x - 4)^2}$ | 20 $\ln \frac{x^2(2x - 1)^3}{(x + 5)^2}$ |
| 21 $\ln \sqrt[3]{\frac{x^2 - 1}{x^2 + 1}}$ | 22 $\ln(\ln x)$ |
| 23 $\ln(x + \sqrt{x^2 - 1})$ | 24 $\ln(x + \sqrt{x^2 + 1})$ |
| 25 $\ln \cos 2x$ | 26 $\cos(\ln 2x)$ |
| 27 $(\sin x)(\ln \sin x)$ | 28 $(\sin 5x)/(\ln x)$ |

In Exercises 29–34 use implicit differentiation to find y' .

- 29 $3y - x^2 + \ln xy = 2$
 30 $y^2 + \ln(x/y) - 4x + 3 = 0$
 31 $x \ln y - y \ln x = 1$

32 $y^3 + x^2 \ln y = 5x + 3$

33 $x \sin y = 1 + y \ln x$

34 $y = \ln(\sin x + \cos y)$

35 Find an equation of the tangent line to the graph of $y = x^2 + \ln(2x - 5)$ at the point on the graph with x -coordinate 3.

36 Find an equation of the tangent line to the graph of $y = x + \ln x$ which is perpendicular to the line $2x + 6y = 5$.

37 What is the difference between the graphs of $y = \ln(x^2)$ and $y = 2 \ln x$?

38 If $0 < a < b$, show that the natural logarithmic function satisfies the hypotheses of the Mean Value Theorem (4.12) on $[a, b]$, and find a general formula for the number c in the conclusion of (4.12).

39 Find the points on the graph of $y = x^2 + 4 \ln x$ at which the tangent line is parallel to the line $y - 6x + 3 = 0$.

40 Find an equation of the tangent line to the graph of $x^3 - x \ln y + y^3 = 2x + 5$ at the point $(2, 1)$.

41 If $\ln 2.00 \approx 0.6932$ use differentials to approximate $\ln 2.01$.

42 If $f(x) = \ln x$, find a formula for the n th derivative $f^{(n)}(x)$, where n is any positive integer.

43 The position function of a point moving on a coordinate line is given by $s(t) = t^2 - 4 \ln(t + 1)$, where $0 \leq t \leq 4$. Find the velocity and acceleration at time t and describe the motion of the point during the time interval $[0, 4]$.

- 44 The equation

$$t = \frac{1}{c(a-b)} \ln \frac{b(a-x)}{a(b-x)}$$

occurs in the study of certain chemical reactions, where x is the concentration of a substance at time t and a , b , c are constants. Prove that $dx/dt = c(a-x)(b-x)$.

- 45 Use Table C or a calculator to help sketch the graph of $y = \ln x$. Find the slope of the tangent line to the graph at the points with x -coordinates 1, 5, 10, 100, and 1000. What is true as the x -coordinate a of the point of tangency increases without bound? What is true if a approaches 0?
- 46 Sketch the graphs of (a) $y = \ln |x|$; (b) $y = |\ln x|$.
- 47 Prove that for every $x > 0$, $(x-1)/x \leq \ln x \leq x-1$.

- 48 Use Exercise 47 to prove that
- $\lim_{x \rightarrow 0} \frac{\ln(x+1)}{x} = 1$
- .

Verify the formulas in Exercises 49–52.

- 49 $\int \ln x \, dx = x \ln x - x + C$
- 50 $\int (\ln x)^2 \, dx = x(\ln x)^2 - 2x \ln x + 2x + C$
- 51 $\int \frac{1}{\sqrt{x^2 + a^2}} \, dx = \ln(x + \sqrt{x^2 + a^2}) + C$
- 52 $\int \frac{1}{a^2 - x^2} \, dx = \frac{1}{2a} \ln \frac{a+x}{a-x} + C, a^2 > x^2$

7.3 The Natural Exponential Function

In Section 7.2 we proved that

$$\lim_{x \rightarrow \infty} \ln x = \infty \quad \text{and} \quad \lim_{x \rightarrow 0^+} \ln x = -\infty.$$

These facts are used in the proof of the following important result.

Theorem (7.7)

To every real number x there corresponds a unique positive real number y such that $\ln y = x$.

Proof First note that if $x = 0$, then $\ln 1 = 0$. Moreover, since $\ln$ is an increasing function, 1 is the only value of y such that $\ln y = 0$.

If x is positive, then we may choose a number b such that

$$\ln 1 < x < \ln b.$$

Since $\ln$ is continuous, the Intermediate Value Theorem (2.29) guarantees the existence of a number y between 1 and b such that $\ln y = x$. Again, since $\ln$ is an increasing function, there is only one such number.

Finally, if x is negative, then there is a $b > 0$ such that

$$\ln b < x < \ln 1$$

and as before, there is precisely one number y between b and 1 such that $\ln y = x$. □

It follows from Theorem (7.7) that the range of $\ln$ is $\mathbb{R}$. Since $\ln$ is an increasing function, it is one-to-one and therefore has an inverse function, to which we give the following special name.

Definition (7.8)

The **natural exponential function**, denoted by **exp**, is the inverse of the natural logarithmic function.

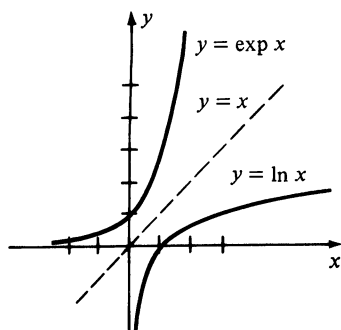


FIGURE 7.6

The reason for the term *exponential* in this definition will become clear shortly. Since $\exp$ is the inverse of $\ln$, its domain is $\mathbb{R}$ and its range is the set of nonnegative real numbers. Moreover, as in (7.2),

$$y = \exp x \quad \text{if and only if} \quad x = \ln y$$

where x is any real number and $y > 0$. By Definition (7.1) we may also write

$$\ln(\exp x) = x \quad \text{and} \quad \exp(\ln y) = y.$$

As pointed out in Section 7.1, if two functions are inverses of one another, then their graphs are reflections through the line $y = x$. Hence the graph of $y = \exp x$ can be obtained by reflecting the graph of $y = \ln x$ through this line, as illustrated in Figure 7.6. Evidently,

$$\lim_{x \rightarrow \infty} \exp x = \infty \quad \text{and} \quad \lim_{x \rightarrow -\infty} \exp x = 0.$$

By Theorem (7.7) there is a unique positive real number whose natural logarithm is 1. This number is denoted by e . The great Swiss mathematician Leonhard Euler (1707–1783) was among the first to study its properties extensively.

Definition of e (7.9)

The letter e denotes the unique positive real number such that $\ln e = 1$.

Several values of $\ln$ were calculated in Section 7.2. It can be shown, by means of the Trapezoidal Rule, that

$$\int_1^{2.7} \frac{1}{t} dt < 1 < \int_1^{2.8} \frac{1}{t} dt$$

(see Exercise 13 in Section 5.7). Consequently, by Definition (7.3),

$$\ln 2.7 < \ln e < \ln 2.8$$

which implies that

$$2.7 < e < 2.8.$$

Later, in Theorem (7.24) it will be shown that e may be expressed as the following limit:

$$e = \lim_{h \rightarrow 0} (1 + h)^{1/h}.$$

This formula can be used to approximate e to any degree of accuracy. In particular, to five decimal places (see page 357),

$$e \approx 2.71828.$$

It can be shown that e is an irrational number.

If r is any rational number, then using (iii) of Theorem (7.6),

$$\ln e^r = r \ln e = r(1) = r.$$

This equation may be used to motivate a definition of e^x , where x is any real number. Specifically, we shall *define* e^x as the real number y such that $\ln y = x$. The following equivalent statement is a convenient way to remember this fact.

Definition of e^x (7.10)

If x is any real number, then

$$e^x = y \quad \text{if and only if} \quad \ln y = x.$$

Recall, from Definition (7.8), that

$$\exp x = y \quad \text{if and only if} \quad \ln y = x.$$

Comparing this relationship with Definition (7.10) we see that

$$e^x = \exp x \quad \text{for every } x.$$

This is the reason for calling $\exp$ an *exponential* function. Indeed, it is often called the **exponential function with base e** . The fact that $\ln(\exp x) = x$ for every x , and $\exp(\ln x) = x$ for every $x > 0$ may now be written as follows.

Theorem (7.11)

$$(i) \quad \ln e^x = x \quad \text{for every } x.$$

$$(ii) \quad e^{\ln x} = x \quad \text{for every } x > 0.$$

Since e^x was not defined as a power of e , it is not obvious that the usual laws of exponents hold. However, these laws may be established easily, as indicated in the proof of the next theorem.

Theorem (7.12)

If p and q are real numbers and r is a rational number, then

$$(i) \quad e^p e^q = e^{p+q}$$

$$(ii) \quad \frac{e^p}{e^q} = e^{p-q}$$

$$(iii) \quad (e^p)^r = e^{pr}.$$

Proof Using Theorems (7.6) and (7.11),

$$\ln e^p e^q = \ln e^p + \ln e^q = p + q = \ln e^{p+q}.$$

Since the natural logarithmic function is one-to-one it follows that

$$e^p e^q = e^{p+q}.$$

This proves (i). The proofs for (ii) and (iii) are similar and are left as an exercise. It will be shown in Section 7.5 that (iii) is also true if r is irrational. $\square$

It is not difficult to show that if x is rational, then e^x has the same meaning as e raised to the power x . For example, we may use Theorem (7.11) as follows:

$$e^0 = e^{\ln 1} = 1$$

$$e^1 = e^{\ln e} = e.$$

Next, from Theorem (7.12),

$$e^2 = e^{1+1} = e^1 e^1 = ee$$

$$e^3 = e^{2+1} = e^2 e^1 = (ee)e$$

and in general, if n is a positive integer, e^n is a product of n factors, all equal to e . Negative exponents also have the usual properties, that is,

$$e^{-1} = e^{0-1} = \frac{e^0}{e^1} = \frac{1}{e}.$$

Similarly,

$$e^{-n} = \frac{1}{e^n}$$

if n is any positive integer. Rational powers of e may also be interpreted as they are in elementary algebra.

The graph of $y = e^x$ is the same as that of $y = \exp x$ illustrated in Figure 7.6. Hereafter we shall use e^x instead of $\exp x$ to denote values of the natural exponential function.

In precalculus mathematics, graphs of equations such as $y = 2^x$ and $y = 3^x$ are sketched by *assuming* that these exponential expressions are defined for all

real x and increase as x increases. Using this intuitive point of view, a rough sketch of the graph of $y = e^x$ can be obtained by sketching $y = (2.7)^x$.

It will be shown in Section 7.7 that the inverse function of a differentiable function is differentiable. Anticipating this result, let us find the derivative of the natural exponential function implicitly. Thus, if

$$y = e^x, \quad \text{then} \quad \ln y = x.$$

Differentiating implicitly gives us

$$D_x(\ln y) = D_x(x), \quad \text{or} \quad \frac{1}{y} D_x y = 1.$$

Multiplying both sides of the last equation by y we obtain

$$D_x y = y.$$

This gives us the following important formula, which is true for every real number x .

Theorem (7.13)

$$D_x e^x = e^x$$

Note that the preceding theorem states that the natural exponential function is its own derivative.

Example 1 Find $f'(x)$ if $f(x) = x^2 e^x$.

Solution By the Product Rule,

$$\begin{aligned} f'(x) &= x^2 D_x e^x + e^x D_x x^2 \\ &= x^2 e^x + e^x (2x) = (x + 2)xe^x. \end{aligned}$$

■

The next result is a generalization of Theorem (7.13).

Theorem (7.14)

If $u = g(x)$ and g is differentiable, then

$$D_x e^u = e^u D_x u.$$

Proof Letting $y = e^u$, $u = g(x)$, and using the Chain Rule,

$$\frac{dy}{dx} = \frac{dy}{du} \frac{du}{dx} = e^u D_x u.$$

□

If $u = x$ then Theorem (7.14) reduces to (7.13).

Example 2 Find y' if $y = e^{\sqrt{x^2+1}}$.

Solution By Theorem (7.14),

$$\begin{aligned} D_x e^{\sqrt{x^2+1}} &= e^{\sqrt{x^2+1}} D_x \sqrt{x^2+1} = e^{\sqrt{x^2+1}} D_x (x^2+1)^{1/2} \\ &= e^{\sqrt{x^2+1}} \left(\frac{1}{2}\right)(x^2+1)^{-1/2}(2x) = e^{\sqrt{x^2+1}} \cdot \frac{x}{\sqrt{x^2+1}} \\ &= \frac{x e^{\sqrt{x^2+1}}}{\sqrt{x^2+1}}. \end{aligned}$$

Example 3 The function f defined by $f(x) = e^{-x^2/2}$ arises in the branch of mathematics called *probability*. Find the local extrema of f , discuss concavity, find the points of inflection, and sketch the graph of f .

Solution By Theorem (7.14),

$$f'(x) = e^{-x^2/2} D_x \left(-\frac{x^2}{2} \right) = e^{-x^2/2} (-2x/2) = -x e^{-x^2/2}.$$

Since $e^{-x^2/2}$ is always positive, the only critical number of f is 0. If $x < 0$, then $f'(x) > 0$ whereas if $x > 0$, then $f'(x) < 0$. It follows from the First Derivative Test that f has a local maximum at 0. The maximum value is $f(0) = e^{-0} = 1$.

Applying the Product Rule to $f'(x)$,

$$\begin{aligned} f''(x) &= -x e^{-x^2/2} (-x) - e^{-x^2/2} \\ &= e^{-x^2/2} (x^2 - 1) \end{aligned}$$

and hence the second derivative is zero at -1 and 1 . If $-1 < x < 1$, then $f''(x) < 0$ and, by (4.16), the graph of f is concave downward in the open interval $(-1, 1)$. If $x < -1$ or $x > 1$, then $f''(x) > 0$ and, therefore, the graph is concave upward throughout the infinite intervals $(-\infty, -1)$ and $(1, \infty)$. Consequently, $P(-1, e^{-1/2})$ and $Q(1, e^{-1/2})$ are points of inflection. Writing

$$f(x) = \frac{1}{e^{x^2/2}}$$

it is evident that as x increases numerically, $f(x)$ approaches 0. It is left as an exercise to prove that $\lim_{x \rightarrow \infty} f(x) = 0$ and $\lim_{x \rightarrow -\infty} f(x) = 0$; that is, the x -axis is a horizontal asymptote. The graph of f is sketched in Figure 7.7.

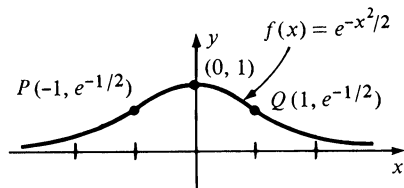


FIGURE 7.7

Exponential functions play an important role in the field of *radiotherapy*, where tumors are treated by means of radiation. Of major interest is the fraction of a tumor population that survives a treatment. This *surviving fraction* depends not only on the energy and nature of the radiation, but also on the

depth, size, and characteristics of the tumor itself. The exposure to radiation may be thought of as a number of potentially damaging events, where only one “hit” is required to kill a tumor cell. Suppose that each cell has exactly one “target” that must be hit. If k denotes the average target size of a tumor cell, and if x is the number of damaging events (the *dose*), then the surviving fraction $f(x)$ is given by

$$f(x) = e^{-kx}.$$

This is called the *one target–one hit surviving fraction*.

Suppose next that each cell has n targets and that hitting any one of the targets results in the death of a cell. In this case, the *n target–one hit surviving fraction* is given by

$$f(x) = 1 - (1 - e^{-kx})^n.$$

The graph of f may be analyzed to determine what effect increasing the dosage x will have on decreasing the surviving fraction of tumor cells. Note that $f(0) = 1$; that is, if there is no dose, then all cells survive. By differentiation, we obtain

$$f'(x) = -nke^{-kx}(1 - e^{-kx})^{n-1}.$$

Let us consider the special case $n = 2$. In this event

$$f'(x) = -2ke^{-kx}(1 - e^{-kx}).$$

Since $f'(x) < 0$ and $f'(0) = 0$, the function f is decreasing and the graph has a horizontal tangent line at the point $(0, 1)$. The reader may verify that the second derivative is given by

$$f''(x) = 2k^2e^{-kx}(1 - 2e^{-kx}).$$

Evidently, $f''(x) = 0$ if $1 - 2e^{-kx} = 0$; that is, if $e^{-kx} = \frac{1}{2}$, or $-kx = \ln \frac{1}{2} = -\ln 2$. This gives us

$$x = \frac{1}{k} \ln 2.$$

It can be shown that if $0 \leq x < (1/k) \ln 2$, then $f''(x) < 0$, and hence the graph is concave downward. If $x > (1/k) \ln 2$ we have $f''(x) > 0$, and the graph is concave upward. This implies that there is a point of inflection with x -coordinate $(1/k) \ln 2$. The y -coordinate of this point is

$$\begin{aligned} f\left(\frac{1}{k} \ln 2\right) &= 1 - (1 - e^{-\ln 2})^2 \\ &= 1 - \left(1 - \frac{1}{2}\right)^2 = \frac{3}{4}. \end{aligned}$$

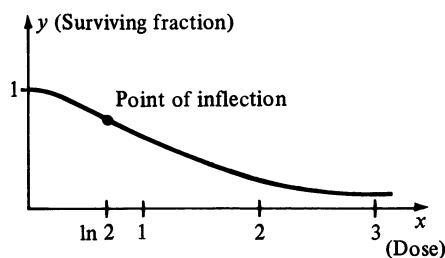


FIGURE 7.8 Surviving fraction of tumor cells after a radiation treatment

The graph is sketched in Figure 7.8 for the case $k = 1$. The “shoulder” on the curve near the point $(0, 1)$ represents the threshold nature of the treatment; that is, a small dose results in very little tumor elimination. Note that for large

x , an increase in dosage has little effect on the surviving fraction. To determine the ideal dose that should be administered to a given patient, specialists in radiation therapy must also take into account the number of healthy cells that are killed during a treatment.

7.3 Exercises

In Exercises 1–30 find $f'(x)$ if $f(x)$ equals the given expression.

- | | |
|--|--|
| 1 e^{-5x} | 2 e^{3x} |
| 3 e^{3x^2} | 4 e^{1-x^3} |
| 5 $\sqrt{1+e^{2x}}$ | 6 $1/(e^x + 1)$ |
| 7 $e^{\sqrt{x}+1}$ | 8 xe^{-x} |
| 9 x^2e^{-2x} | 10 $\sqrt{e^{2x} + 2x}$ |
| 11 $e^x/(x^2 + 1)$ | 12 x/e^{x^2} |
| 13 $(e^{4x} - 5)^3$ | 14 $(e^{3x} - e^{-3x})^4$ |
| 15 $e^{1/x} + 1/e^x$ | 16 $e^{\sqrt{x}} + \sqrt{e^x}$ |
| 17 $\frac{e^x - e^{-x}}{e^x + e^{-x}}$ | 18 $e^{(e^x)}$ |
| 19 $e^{-2x} \ln x$ | 20 $e^{x \ln x}$ |
| 21 $\ln \frac{e^x + 1}{e^x - 1}$ | 22 $\frac{\ln(e^x + 1)}{\ln(e^x - 1)}$ |
| 23 $\sqrt{\ln(e^{2x} + e^{-2x})}$ | 24 $\ln \sqrt{e^{2x} + e^{-2x}}$ |
| 25 $e^{\ln x}$ | 26 $\ln e^x$ |
| 27 $\sin e^{5x}$ | 28 $e^{\sin 5x}$ |
| 29 $\ln \cos e^{-x}$ | 30 $e^{-3x} \cos 3x$ |

In Exercises 31–34 use implicit differentiation to find y' .

- 31 $e^{xy} - x^3 + 3y^2 = 11$
- 32 $xe^y + 2x - \ln(y + 1) = 3$
- 33 $y^3 + xe^y = 3x^2 - 10$
- 34 $e^x \cos y = xe^y$
- 35 Find an equation of the tangent line to the graph of $y = (x - 1)e^x + 3 \ln x + 2$ at the point $P(1, 2)$.
- 36 Find an equation of the tangent line to the graph of $y = x - e^{-x}$ that is parallel to the line $6x - 2y = 7$.
- 37 Show that the natural exponential function satisfies the hypotheses of the Mean Value Theorem (4.12) on every closed interval $[a, b]$, and find a general formula for the number c in the conclusion of (4.12).

- 38 Compare the graphs of $y = \sqrt{e^x}$ and $y = e^{\sqrt{x}}$.

- 39 Find a point on the graph of $y = e^{2x}$ at which the tangent line passes through the origin.
- 40 Find an equation of the normal line to the graph of $y = 4xe^{x^2-1}$ at the point $P(1, 4)$.

In Exercises 41–46 find the local extrema of f . Determine where f is increasing or decreasing, discuss concavity, find the points of inflection, and sketch the graph of f .

- 41 $f(x) = xe^x$
- 42 $f(x) = x^2e^{-2x}$
- 43 $f(x) = e^{-2x}$
- 44 $f(x) = xe^{-x}$
- 45 $f(x) = x \ln x$
- 46 $f(x) = e^{-x} \sin x$, where $0 \leq x \leq 4\pi$
- 47 A radioactive substance decays according to the law $q(t) = q_0 e^{-ct}$, where q_0 is the initial amount of the substance, c is a positive constant, and $q(t)$ is the amount remaining after t units of time. Show that the rate at which the substance decays is proportional to $q(t)$.
- 48 The current $I(t)$ at time t in a certain electrical circuit is given by $I(t) = I_0 e^{-Rt/L}$, where R and L denote the resistance and inductance, respectively, and I_0 is the current at time $t = 0$. Show that the rate of change of the current at any time t is proportional to $I(t)$.
- 49 If a drug is injected into the blood stream, then its concentration t minutes later is given by

$$C(t) = \frac{k}{a-b} (e^{-bt} - e^{-at})$$

where a , b and k are positive constants.

- (a) At what time does the maximum concentration occur?
- (b) What can be said about the concentration after a long period of time? (Justify your answer.)

- 50 If a beam of light that has intensity k is projected vertically downward into water, then its intensity $I(x)$ at a depth of x meters is $I(x) = ke^{-1.4x}$.
- (a) At what rate is the intensity changing with respect to depth at 1 meter? 5 meters? 10 meters?
- (b) At what depth is the intensity one-half its value at the surface? One-tenth its value?
- 51 Use differentials to approximate the change in $f(x) = xe^{x^2}$ if x changes from 1.00 to 1.01. What is the approximate value of $f(1.01)$?
- 52 If $f(x) = e^{2x}$, find a formula for the n th derivative $f^{(n)}(x)$.

In Exercises 53–55 find values of c such that $y = e^{cx}$ is a solution of the given differential equation.

- 53 $y'' - 3y' + 2y = 0$
- 54 $y'' - 9y = 0$
- 55 $y''' - y'' - 4y' + 4y = 0$
- 56 Prove that $y = ae^{-x} \cos 2x + be^{-x} \sin 2x$, where a and b are constants, is a solution of the differential equation $y'' + 2y' + 5y = 0$.
- 57 Prove the following.
- (a) $\lim_{x \rightarrow \infty} e^{-x^2/2} = 0$ (b) $\lim_{x \rightarrow -\infty} e^{-x^2/2} = 0$
- 58 Prove the following.
- (a) $\lim_{x \rightarrow -\infty} e^x = 0$ (b) $\lim_{x \rightarrow \infty} e^x = \infty$
- 59 In statistics the *normal distribution function* f is defined by

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{(-1/2)[(x-\mu)/\sigma]^2}$$

where μ and σ are constants, and $\sigma > 0$. (μ is called the *mean* and σ the *variance* of the distribution.) Find the

local extrema of f and determine where f is increasing or decreasing. Discuss concavity, find points of inflection, find $\lim_{x \rightarrow \infty} f(x)$ and $\lim_{x \rightarrow -\infty} f(x)$, and sketch the graph of f .

- 60 Prove that $e^x \geq 1 + x$ for all x . (Hint: Use the Mean Value Theorem.)
- 61 Prove (ii) and (iii) of Theorem (7.12).
- 62 The integral $\int_a^b e^{-x^2} dx$ has important applications. Use the Trapezoidal Rule (with $n = 10$) and a calculator to approximate this integral if $a = 0$ and $b = 1$.
- 63 Nerve impulses in the human body travel along nerve fibers that consist of an *axon*, which transports the impulse, and an insulating coating surrounding the axon called the *myelin sheath* (see figure). *Nodes of Ranvier* serve as relay stations along the fibers; however, between nodes the nerve fiber is similar to an insulated cylindrical cable. It is known that for such cables, the velocity v of an impulse is given by $v = -k(r/R)^2 \ln(r/R)$ where r is the radius of the cable and R is the insulation radius. Find the value of r/R that maximizes v . (In most nerve fibers $r/R \approx 0.6$.)

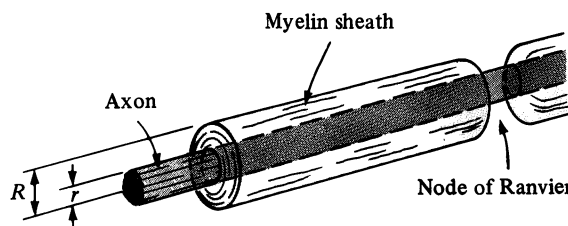


FIGURE FOR EXERCISE 63

- 64 Sketch the graph of the *three target-one hit surviving fraction* (with $k = 1$) given by $f(x) = 1 - (1 - e^{-x})^3$. Compare the graph with Figure 7.8.

7.4 Differentiation and Integration

The formula for the derivative of $\ln u$ established in Section 7.2 can be extended as indicated in the next theorem.

Theorem (7.15)

If $u = g(x)$, where g is differentiable and $g(x) \neq 0$, then

$$D_x \ln |u| = \frac{1}{u} D_x u.$$

Proof If $x < 0$, then $\ln |x| = \ln (-x)$ and by Theorem (7.5),

$$D_x \ln (-x) = \frac{1}{(-x)} D_x(-x) = \frac{1}{(-x)} (-1) = \frac{1}{x}.$$

Consequently,

$$D_x \ln |x| = \frac{1}{x} \quad \text{for all } x \neq 0.$$

The Chain Rule may now be used to complete the proof. □

Observe that if $u = g(x) > 0$, then Theorem (7.15) is the same as (7.5).

Example 1 Find $f'(x)$ if $f(x) = \ln |4 + 5x - 2x^3|$.

Solution Using Theorem (7.15) with $u = 4 + 5x - 2x^3$,

$$f'(x) = \frac{1}{4 + 5x - 2x^3} (5 - 6x^2) = \frac{5 - 6x^2}{4 + 5x - 2x^3}. \quad \blacksquare$$

We may use differentiation formulas for $\ln$ to obtain rules for integration. In particular, since

$$D_x \ln |g(x)| = \frac{1}{g(x)} g'(x)$$

the following integration formula holds.

Theorem (7.16)

If $u = g(x) \neq 0$, and g is differentiable,

$$\int \frac{1}{u} du = \ln |u| + C.$$

Of course, if $u > 0$, then the absolute value sign may be deleted. A special case of Theorem (7.16) is

$$\int \frac{1}{x} dx = \ln |x| + C.$$

Example 2 Find $\int \frac{x}{3x^2 - 5} dx$.

Solution If we let $u = 3x^2 - 5$, then $du = 6x \, dx$. We may arrive at the form (7.16) by introducing a factor 6 in the integrand as follows:

$$\begin{aligned}\int \frac{x}{3x^2 - 5} \, dx &= \frac{1}{6} \int \frac{1}{3x^2 - 5} 6x \, dx = \frac{1}{6} \int \frac{1}{u} \, du \\ &= \frac{1}{6} \ln |u| + C = \frac{1}{6} \ln |3x^2 - 5| + C.\end{aligned}$$

Another technique is to replace the expression $x \, dx$ in the integral by $\frac{1}{6} du$ and then integrate. ■

Example 3 Evaluate $\int_2^4 \frac{1}{9 - 2x} \, dx$.

Solution Since $1/(9 - 2x)$ is continuous on $[2, 4]$, the definite integral exists. One method of evaluation consists of using an indefinite integral to find an antiderivative of $1/(9 - 2x)$. If we let $u = 9 - 2x$, then $du = -2 \, dx$ and we may proceed as follows:

$$\begin{aligned}\int \frac{1}{9 - 2x} \, dx &= -\frac{1}{2} \int \frac{1}{9 - 2x} (-2) \, dx \\ &= -\frac{1}{2} \int \frac{1}{u} \, du = -\frac{1}{2} \ln |u| + C \\ &= -\frac{1}{2} \ln |9 - 2x| + C.\end{aligned}$$

Applying the Fundamental Theorem of Calculus,

$$\begin{aligned}\int_2^4 \frac{1}{9 - 2x} \, dx &= -\frac{1}{2} \left[\ln |9 - 2x| \right]_2^4 \\ &= -\frac{1}{2} [\ln 1 - \ln 5] = \frac{1}{2} \ln 5.\end{aligned}$$

Another method is to use the same substitution in the *definite* integral and change the limits of integration. Referring to $u = 9 - 2x$ we see that if $x = 2$ then $u = 5$, and if $x = 4$ then $u = 1$. Consequently,

$$\begin{aligned}\int_2^4 \frac{1}{9 - 2x} \, dx &= -\frac{1}{2} \int_2^4 \frac{1}{9 - 2x} (-2) \, dx \\ &= -\frac{1}{2} \int_5^1 \frac{1}{u} \, du = -\frac{1}{2} \left[\ln |u| \right]_5^1 \\ &= -\frac{1}{2} [\ln 1 - \ln 5] = \frac{1}{2} \ln 5. \quad \blacksquare\end{aligned}$$

Example 4 Find $\int \frac{\sqrt{\ln x}}{x} dx$.

Solution If we let $u = \ln x$, then $du = (1/x) dx$ and

$$\begin{aligned}\int \frac{\sqrt{\ln x}}{x} dx &= \int \sqrt{\ln x} \cdot \frac{1}{x} dx = \int u^{1/2} du = \frac{2}{3} u^{3/2} + C \\ &= \frac{2}{3} (\ln x)^{3/2} + C.\end{aligned}$$

■

It is easy to find integration formulas for the natural exponential function. Since $D_x e^{g(x)} = e^{g(x)} g'(x)$ we obtain the following.

Theorem (7.17)

If $u = g(x)$, and g is differentiable,

$$\int e^u du = e^u + C.$$

As a special case of Theorem (7.17),

$$\int e^x dx = e^x + C.$$

Example 5 Find $\int \frac{e^{3/x}}{x^2} dx$.

Solution If we let $u = 3/x$, then $du = (-3/x^2) dx$ and the integrand may be written in the form (7.17) by introducing the factor -3 . Doing this and compensating by multiplying the integral by $-\frac{1}{3}$,

$$\begin{aligned}\int \frac{e^{3/x}}{x^2} dx &= -\frac{1}{3} \int e^{3/x} \left(-\frac{3}{x^2}\right) dx \\ &= -\frac{1}{3} \int e^u du \\ &= -\frac{1}{3} e^u + C \\ &= -\frac{1}{3} e^{3/x} + C.\end{aligned}$$

■

Example 6 Find $\int_1^2 \frac{e^{3/x}}{x^2} dx$.

Solution An antiderivative was found in Example 5. Applying the Fundamental Theorem of Calculus,

$$\begin{aligned}\int_1^2 \frac{e^{3/x}}{x^2} dx &= -\frac{1}{3} \left[e^{3/x} \right]_1^2 \\ &= -\frac{1}{3} (e^{3/2} - e^3).\end{aligned}$$

This example can also be solved by using the method of substitution. If, as in Example 5 we let $u = 3/x$, then $du = -3/x^2$. Next we note that if $x = 1$, then $u = 3$, and if $x = 2$, then $u = \frac{3}{2}$. Consequently,

$$\begin{aligned}\int_1^2 \frac{e^{3/x}}{x^2} dx &= -\frac{1}{3} \int_3^{3/2} e^u du \\ &= -\frac{1}{3} \left[e^u \right]_3^{3/2} = \left(-\frac{1}{3} \right) (e^{3/2} - e^3).\end{aligned}$$

Example 7 Find the area of the region bounded by the graphs of the equations $y = e^x$, $y = \sqrt{x}$, $x = 0$, and $x = 1$.

Solution The region and a typical rectangle of the type considered in Chapter 6 are shown in Figure 7.9. The area A is

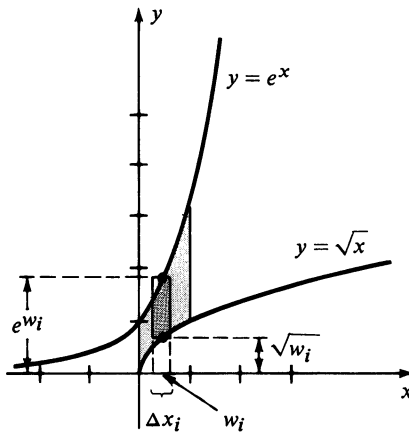


FIGURE 7.9

$$\begin{aligned}A &= \lim_{\|P\| \rightarrow 0} \sum_i (e^{w_i} - \sqrt{w_i}) \Delta x_i \\ &= \int_0^1 (e^x - \sqrt{x}) dx \\ &= \left[e^x - \frac{2}{3} x^{3/2} \right]_0^1 \\ &= \left(e - \frac{2}{3} \right) - (e^0 - 0) = e - \frac{5}{3}.\end{aligned}$$

If an approximation is desired, then to the nearest hundredth,

$$A \approx 2.72 - 1.67 = 1.05.$$

Given $y = f(x)$ it is sometimes convenient to find $D_x y$ by a process called **logarithmic differentiation**. This method is especially useful if $f(x)$ involves complicated products, quotients, or powers. Later in the text we shall also apply it to expressions such as $y = (x^2 + 1)^x$. The process may be outlined as follows where, at the outset, it is assumed that $f(x) > 0$.